

0	0	1	1	1	0	0	1	1	1	0	0	0	1	0	0
0	1	0	0	0	1	1	0	0	0	1	0	1	0	1	0
1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1
1	0	0	0	0	0	0	1	0	0	0	1	1	0	0	1
0	1	1	1	1	1	0	0	1	1	1	1	1	0	0	1
0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	1
1	0	0	0	0	0	1	1	0	0	0	1	1	0	0	1
0	1	1	1	1	1	0	0	1	1	1	0	1	0	0	1

SGALAB

Evolution with the world

SxLAB Family Member

What is Genetic Algorithms ?

à à à Genetic Algorithms(GAs) Brief
& GAs Toolboxes for Matlab intro.

Yi Chen

16-Jan-2009

leo.chen.yi@gmail.com

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What is Genetic Algorithms ?

à Genetic Algorithms(GAs) was formally introduced in the United States in the 1970s by John Holland at University of Michigan.

(John Holland, from the University of Michigan began his work on genetic algorithms at the beginning of the 60s. A first achievement was the publication of *Adaptation in Natural and Artificial System* in 1975.)

"Genetic Algorithms are based on a biological metaphor: They view learning as a competition among a population of evolving candidate problem solutions. A 'fitness' function evaluates each solution to decide whether it will contribute to the next generation of solutions. Then, through operations analogous to gene transfer in sexual reproduction, the algorithm creates a new population of candidate solutions."

--From *Artificial Intelligence, Structures and Strategies for Complex Problem Solving*, Fourth Edition, at page 471. Luger, George F. 2002. Harlow, England: Addison-Wesley.



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FACULTY PROFILE — JOHN HOLLAND

<http://www.lsa.umich.edu/psych/people/directory/profiles/faculty/?uniquename=jholland>

Professor of Psychology and Electrical Engineering & Computer Science

Ph.D. University of Michigan

Area: [Cognition & Perception](#)

Contact Information

Office: 1255 East Hall

Phone: 763-3648

Email: jholland@umich.edu



Research and Teaching Interests

Study of cognitive processes and complex adaptive systems in general, using mathematical models and computer simulation.

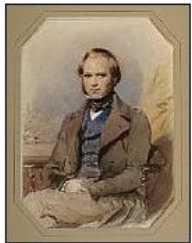
Related Links

- [Dept. of Electrical Engineering & Computer Science](#)
- [EECS Faculty Profile](#)

<http://www.lsa.umich.edu/psych/people/directory/profiles/faculty/?uniquename=jholland>

What is Genetic Algorithms?

Genetic Algorithms is inspired by the theory of Charles Darwin (1809 - 1882) about Evolution.



Richmond - Charles Darwin J980057



Charles Darwin K980352



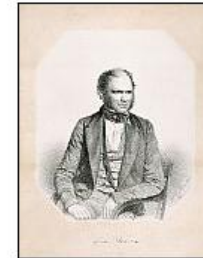
Charles Darwin K970247



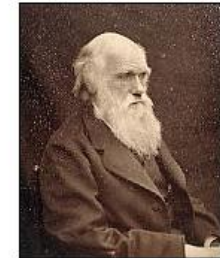
Charles Darwin on the verandah at Down House K970226



Charles Darwin K970234



Charles Darwin K970239



Charles Darwin K970215



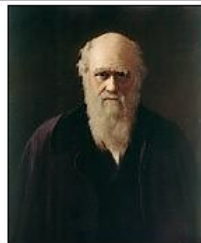
Charles Darwin K980123



Charles Darwin on horseback K970217



Laurence - Charles Darwin J970202



Reilly - Charles Darwin J970164



Evstafieff - Darwin in his Study J970178



Charles Darwin and his son N990002



Charles Darwin K970235



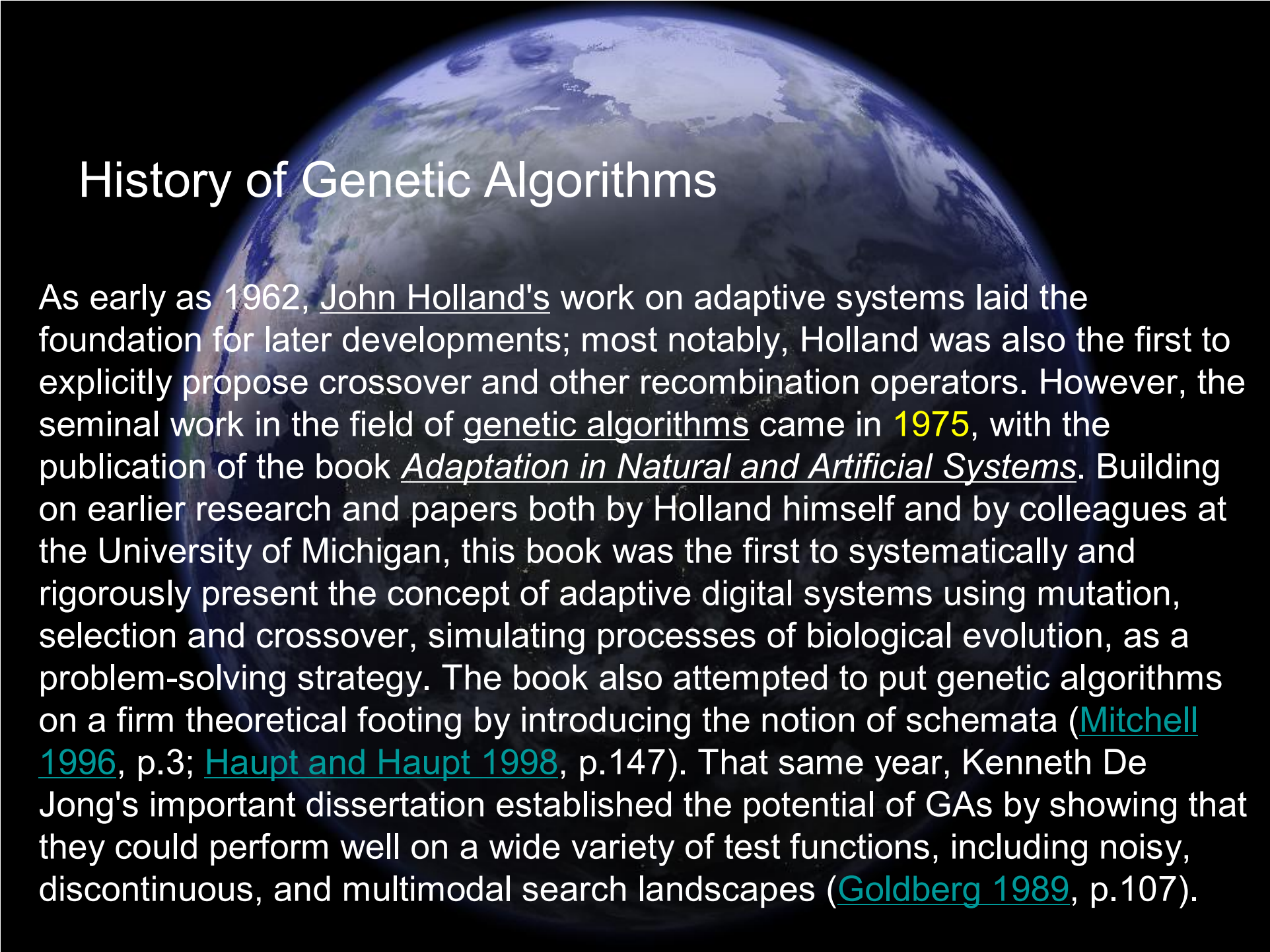
Sharples - Charles Darwin (aged six) and Catherine K971925



www.bbc.co.uk/history/historic_figures/darwin_charles.shtml
www.bbc.co.uk/darwin/
www.darwin200.org/
www.amnh.org/exhibitions/darwin/
www.englishheritageimages.com/

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History of Genetic Algorithms

As early as 1962, John Holland's work on adaptive systems laid the foundation for later developments; most notably, Holland was also the first to explicitly propose crossover and other recombination operators. However, the seminal work in the field of genetic algorithms came in **1975**, with the publication of the book *Adaptation in Natural and Artificial Systems*. Building on earlier research and papers both by Holland himself and by colleagues at the University of Michigan, this book was the first to systematically and rigorously present the concept of adaptive digital systems using mutation, selection and crossover, simulating processes of biological evolution, as a problem-solving strategy. The book also attempted to put genetic algorithms on a firm theoretical footing by introducing the notion of schemata (Mitchell 1996, p.3; Haupt and Haupt 1998, p.147). That same year, Kenneth De Jong's important dissertation established the potential of GAs by showing that they could perform well on a wide variety of test functions, including noisy, discontinuous, and multimodal search landscapes (Goldberg 1989, p.107).

Evolutionary Computation

- à Idea of evolutionary computing was introduced in the 1960s by I. Rechenberg in his work "Evolution Strategies" (ES). His idea was then developed by other researchers.
- à Genetic Algorithms (GAs) were invented by John Holland and developed by him and his students and colleagues. This led to Holland's book "Adaption in Natural and Artificial Systems" published in 1975.
- à In 1992 John Koza has used genetic algorithm to evolve programs to perform certain tasks. He called his method "Genetic programming" (GP). LISP programs were used, because programs in this language can be expressed in the form of a "parse tree", which is the object the GA works on.
- à Evolutionary Programming (EP), originally conceived by Lawrence J. Fogel in 1960, is a stochastic OPTIMIZATION strategy similar to Genetic Algorithms, but instead places emphasis on the behavioral linkage between PARENTs and their OFFSPRING, rather than seeking to emulate specific GENETIC OPERATORs as observed in nature. Evolutionary programming is similar to EVOLUTION STRATEGIES, although the two approaches developed independently
- P.s. Briefly, Computational Intelligence = Genetic Algorithms + Artificial Neural Network + Fuzzy Logic system

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Genetic Algorithms Basic workflow

Max : $F(x_i) = \{ f_1(x_i), f_2(x_i), \dots, f_k(x_i) \}$
 Subject to: $g_j(x_i) \leq 0$

- (1) Object design based on engineering application
 (2) Variable/Fitness function design
- $i = 1, 2, \dots, m$
 $j = 1, 2, \dots, n$

Start

initial solution

population

$x(1)$ $x(2)$ $x(n)$

(10 4554)(74545 4)...(5 4345)
 (11 0230)(75690 3)...(5 2344)
 (9 99234)(74545 9)...(5 2322)
 ...
 (10 4544)(74544 3)...(5 123)

population

$x(1)$ $x(2)$ $x(n)$
 coding_length(1) coding_length(2) coding_length(n)

10001010101010100...0101
 01010000010100101...1101
 01000101000101010...1110
 ...
 00101000101010101...1000

Fitness/Cost Evaluation

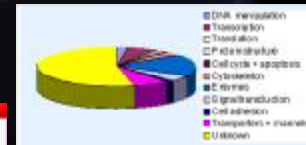
No



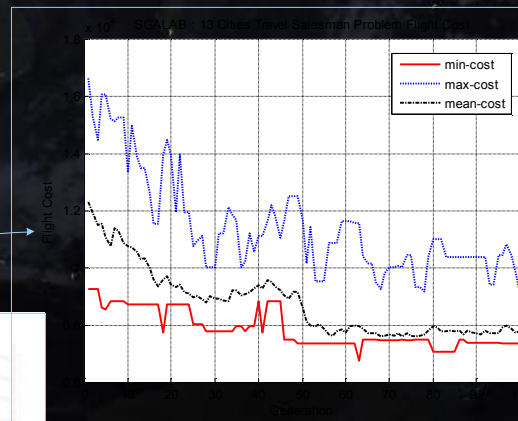
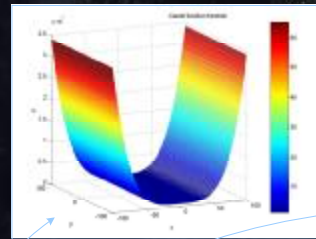
En-coding

Selection

Crossover



Yes



population

$x(1)$ $x(2)$ $x(n)$
 coding_length(1) coding_length(2) coding_length(n)

10001010101000101...1101
 01010000010110100...0101
 00101000101101010...1110
 ...
 01000101000010101...1000

Mutation

population

$x(1)$ $x(2)$ $x(n)$
 coding_length(1) coding_length(2) coding_length(n)

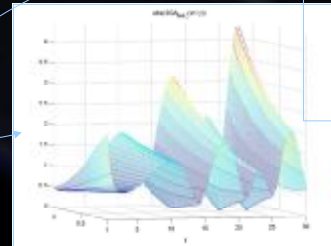
10001010111000101...1101
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 00101000011101010...1100
 ...
 01000101010010001...1000

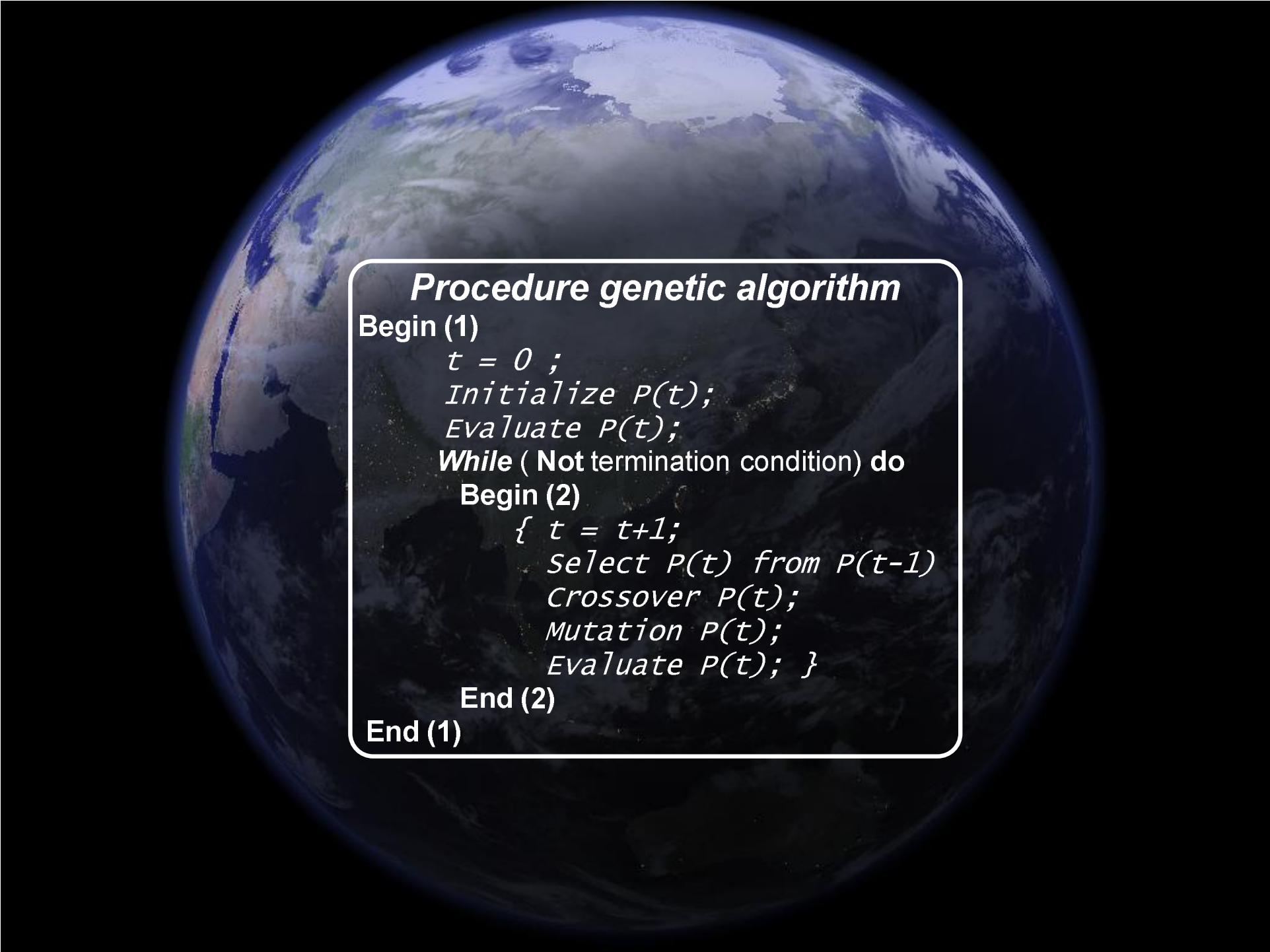
De-coding

Stop



Result & Analysis





Procedure genetic algorithm

Begin (1)

$t = 0$;

Initialize $P(t)$;

Evaluate $P(t)$;

While (Not termination condition) **do**

Begin (2)

{ $t = t+1$;

Select $P(t)$ from $P(t-1)$

Crossover $P(t)$;

Mutation $P(t)$;

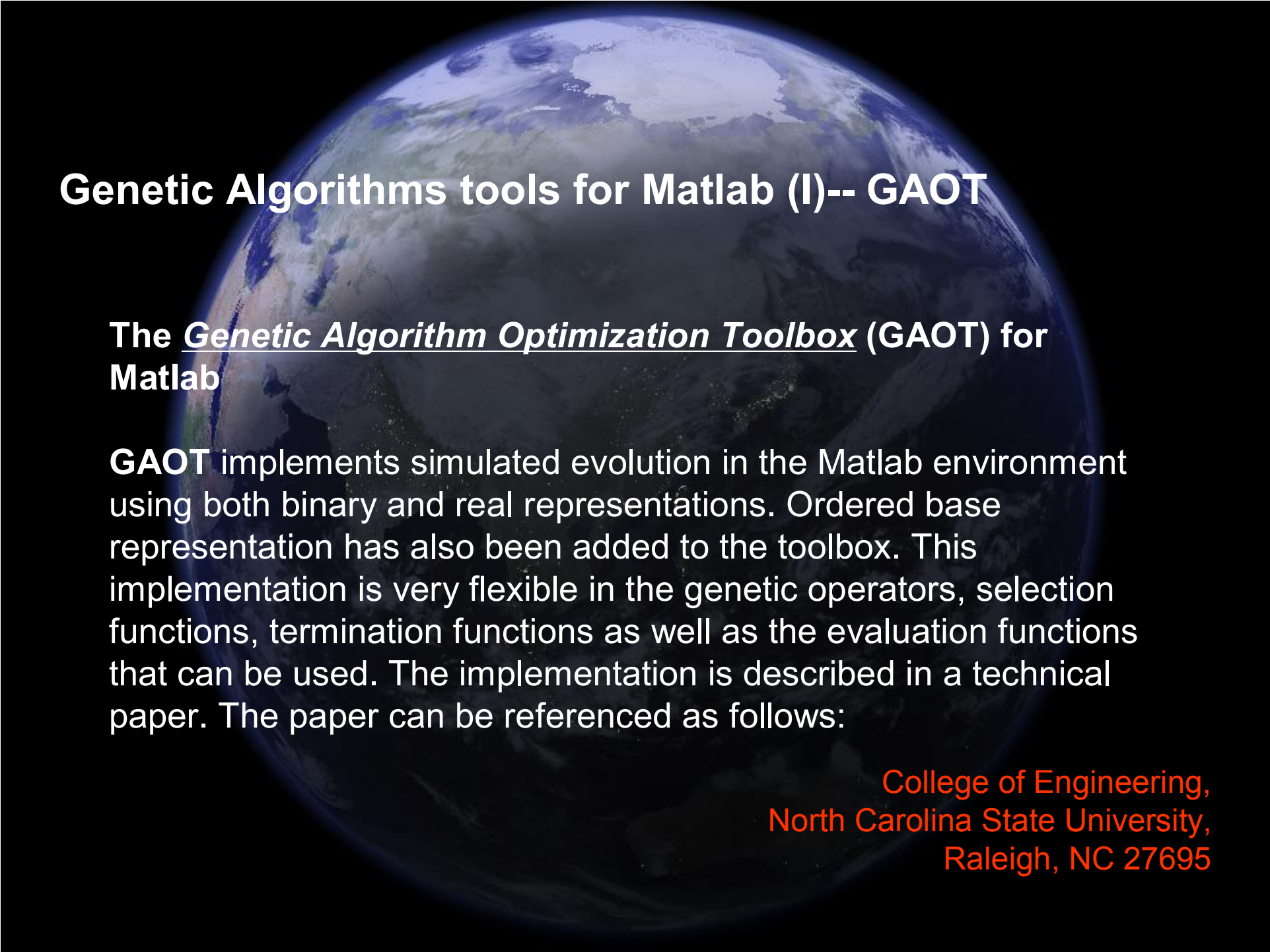
Evaluate $P(t)$; }

End (2)

End (1)

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Genetic Algorithms tools for Matlab (I)-- GAOT

The Genetic Algorithm Optimization Toolbox (GAOT) for Matlab

GAOT implements simulated evolution in the Matlab environment using both binary and real representations. Ordered base representation has also been added to the toolbox. This implementation is very flexible in the genetic operators, selection functions, termination functions as well as the evaluation functions that can be used. The implementation is described in a technical paper. The paper can be referenced as follows:

College of Engineering,
North Carolina State University,
Raleigh, NC 27695



Where can we find GOAT:

<http://www.ie.ncsu.edu/mirage/GAToolBox/gaot/>

The Genetic Algorithm Optimization Toolbox (GAOT) for Matlab 5

GAOT implements simulated evolution in the Matlab environment using both binary and real representations. Ordered base representation has also been added to the toolbox. This implementation is very flexible in the genetic operators, selection functions, termination functions as well as the evaluation functions that can be used. The implementation is described in a technical paper. The paper can be referenced as follows

A Genetic Algorithm for Function Optimization: A Matlab Implementation by Chris Houck, Jeff Joines, and Mike Kay, NCSU-IE TR 95-09, 1995.

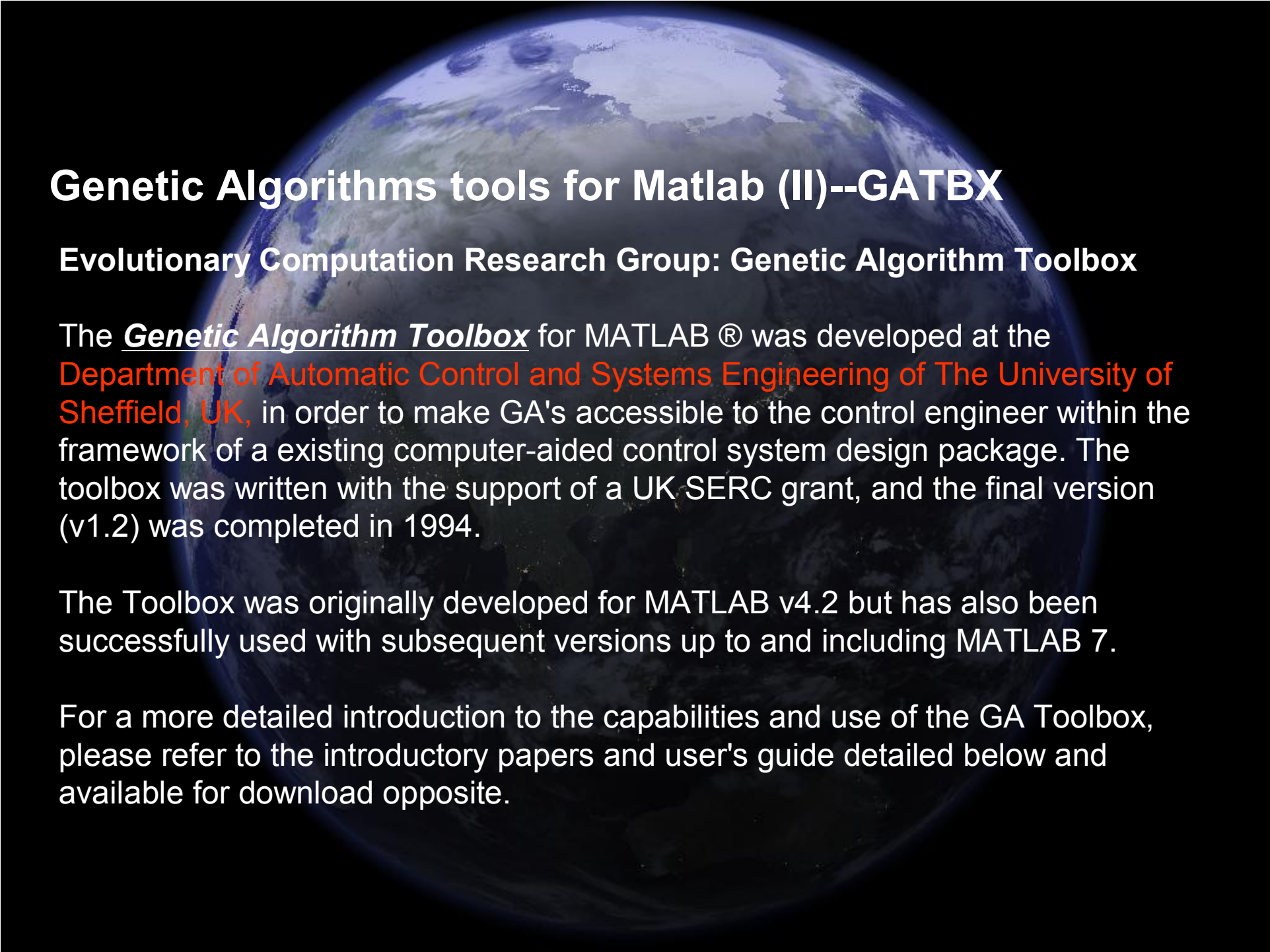
The entire toolbox can be download either as a compressed tar archive ([GAOT.tar.gz](#)) or a ZIP file ([GAOT.zip](#)). This includes the postscript and dvi versions of the companion paper.

The GA toolbox can be also obtained via anonymous ftp from the following directory: <ftp://ftp.eos.ncsu.edu/pub/simul/GAOT> as well as other [GA related papers](#).



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Genetic Algorithms tools for Matlab (II)--GATBX

Evolutionary Computation Research Group: Genetic Algorithm Toolbox

The *Genetic Algorithm Toolbox* for MATLAB ® was developed at the **Department of Automatic Control and Systems Engineering of The University of Sheffield, UK**, in order to make GA's accessible to the control engineer within the framework of a existing computer-aided control system design package. The toolbox was written with the support of a UK SERC grant, and the final version (v1.2) was completed in 1994.

The Toolbox was originally developed for MATLAB v4.2 but has also been successfully used with subsequent versions up to and including MATLAB 7.

For a more detailed introduction to the capabilities and use of the GA Toolbox, please refer to the introductory papers and user's guide detailed below and available for download opposite.

Where can find GATBX:

<http://www.shef.ac.uk/acse/research/ecrg/getgat.html>



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Obtaining the toolbox

The Genetic Algorithm Toolbox for [MATLAB](#)® was developed at the [Department of Automatic Control and Systems Engineering](#) of The University of Sheffield, UK. It was originally developed for MATLAB v4.2, but has also been successfully used with subsequent versions up to and including MATLAB 7.

The GA Toolbox is copyright the original authors and The University of Sheffield, and is published here under the [GNU General Public License](#).

We would be interested to hear of your experiences with, criticisms of, and enhancements to the GA Toolbox. Please direct all such correspondence to ga-toolbox@acse.sheffield.ac.uk.

The GA Toolbox v1.2 is available in two packages: one suitable for DOS/Windows systems, and one suitable for Unix systems.

To download a copy of the Toolbox use the form below

Genetic Algorithm TOOLBOX

For Use with MATLAB®

Andrew Chipperfield
Peter Fleming
Hartmut Pohlheim
Carlos Fonseca

Version 1.2

User's Guide



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Genetic Algorithms tools for Matlab(III) – GADS (Matlab 7.xx)

The *Genetic Algorithm and Direct Search* Toolbox is a collection of functions that extend the capabilities of the Optimization Toolbox and the MATLAB® numeric computing environment. The Genetic Algorithm and Direct Search Toolbox includes routines for solving optimization problems using:

Genetic algorithm

Direct search

These algorithms enable you to solve a variety of optimization problems that lie outside the scope of the standard Optimization Toolbox.

All the toolbox functions are MATLAB M-files, made up of MATLAB statements that implement specialized optimization algorithms. You can view the MATLAB code for these functions using the statement

You can extend the capabilities of the Genetic Algorithm and Direct Search Toolbox by writing your own M-files, or by using the toolbox in combination with other toolboxes, or with MATLAB or Simulink®.

Keyin “gatool” in command window

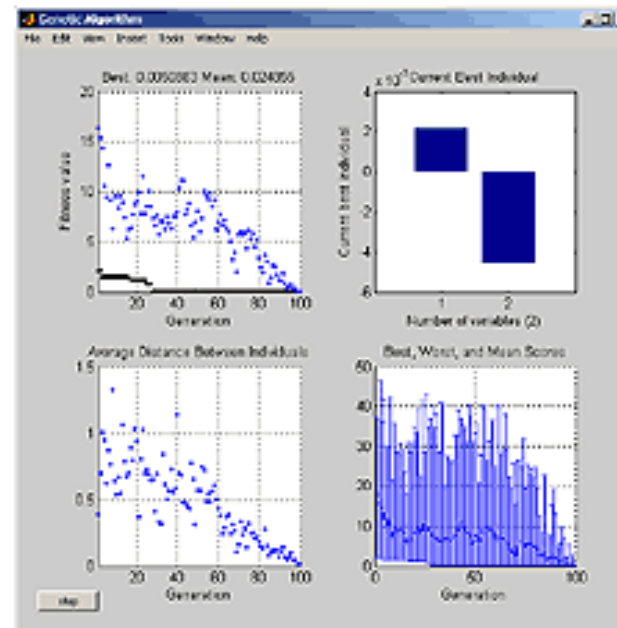
<http://www.mathworks.com/products/gads/>

<http://www.mathworks.com/access/helpdesk/help/toolbox/gads/>

Genetic Algorithm and Direct Search Toolbox 1.0.3

Solve optimization problems using genetic and direct search algorithms

The Genetic Algorithm and Direct Search Toolbox extends the optimization capabilities in MATLAB and the Optimization Toolbox with tools for using the genetic and direct search algorithms. You can use these algorithms for problems that are difficult to solve with traditional optimization techniques, including problems that are not well defined or are difficult to model mathematically. You can also use them when computation of the objective function is discontinuous, highly nonlinear, stochastic, or has unreliable or undefined derivatives.



- [Overview and Key Features](#)
- [Graphical Interface and Command-Line Functions](#)
- [Genetic Algorithm Tools](#)
- [Direct Search Algorithm Tools](#)
- [Using Other Functions and Solvers](#)
- [Displaying, Monitoring, and Outputting Results](#)

Click to display descriptions of options.

Enter fitness function.

Enter number of variables
for the fitness function.

Start the genetic
algorithm.

Results displayed here.

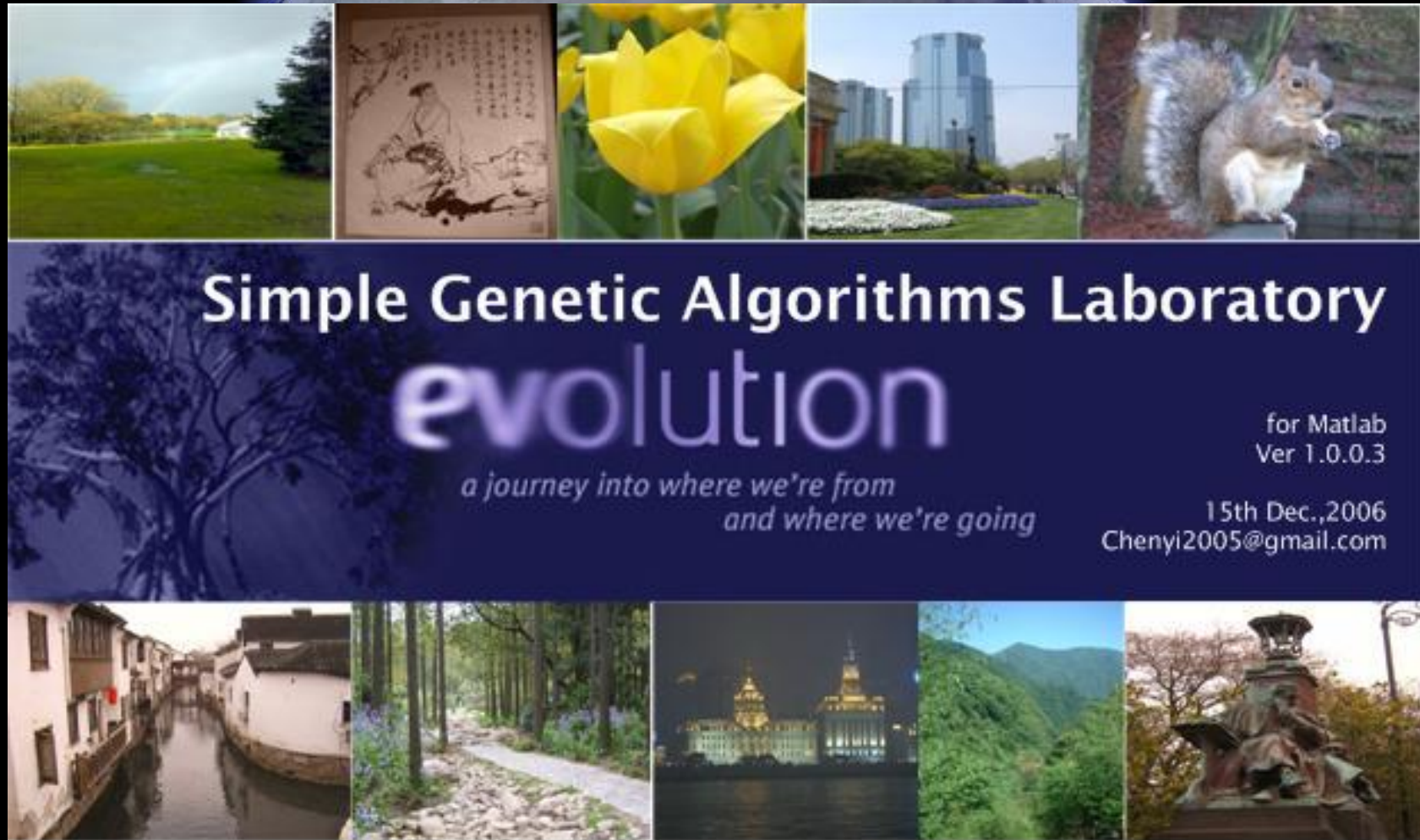
The screenshot shows the 'Genetic Algorithm Tool' window. It has a menu bar with 'File' and 'Help'. The main area is divided into several sections:

- Fitness function:** A text input field with an annotation 'Enter fitness function.' pointing to it.
- Number of variables:** A text input field with an annotation 'Enter number of variables for the fitness function.' pointing to it.
- Plots:** A section with a 'Plot interval' set to 1 and a grid of checkboxes for various plots: Best fitness, Best individual, Distance, Expectation, Genealogy, Range, Score diversity, Scores, Selection, Stopping, and Custom function.
- Run solver:** A section with a checkbox 'Use random states from previous run', three buttons 'Start', 'Pause', and 'Stop', a 'Current generation' input field, and a 'Status and results' text area with an annotation 'Results displayed here.' pointing to it.
- Options:** A sidebar on the right with a '>>' button at the top (annotated 'Click to display descriptions of options.') and a list of expandable options: Population, Fitness scaling, Selection, Reproduction, Mutation, Crossover, Migration, Hybrid function, Stopping criteria, Output function, Display to command window, and Vectorize.
- Initial population:** A text input field with '[]'.
- Initial scores:** A text input field with '[]'.
- Initial range:** A text input field with '[0 ; 1]'.
- Export to Workspace...** A button at the bottom.

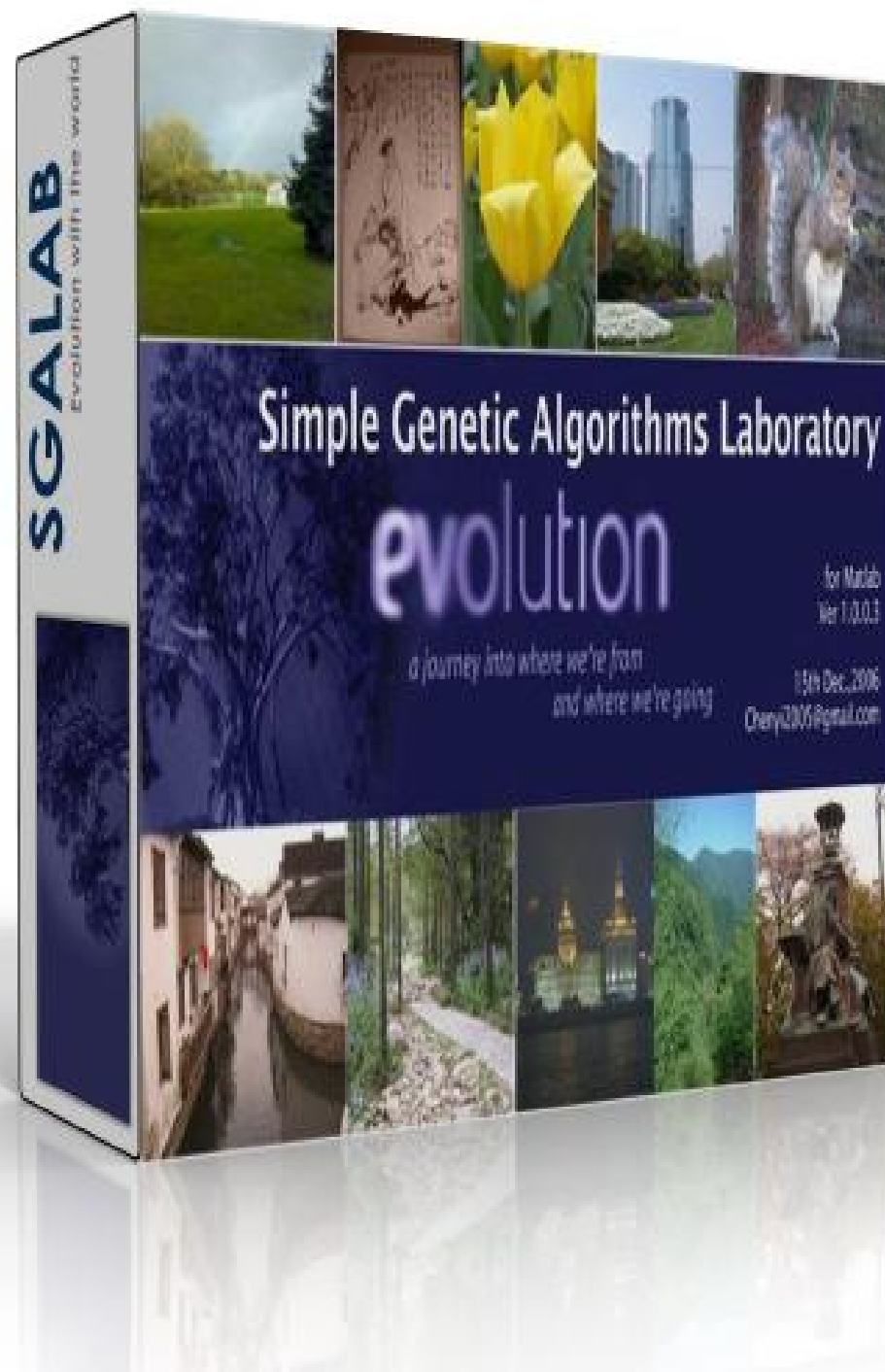
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Genetic Algorithms tools for Matlab(IV) – SGALAB



<http://www.mathworks.com/matlabcentral/fileexchange/loadFile.do?objectId=5882>



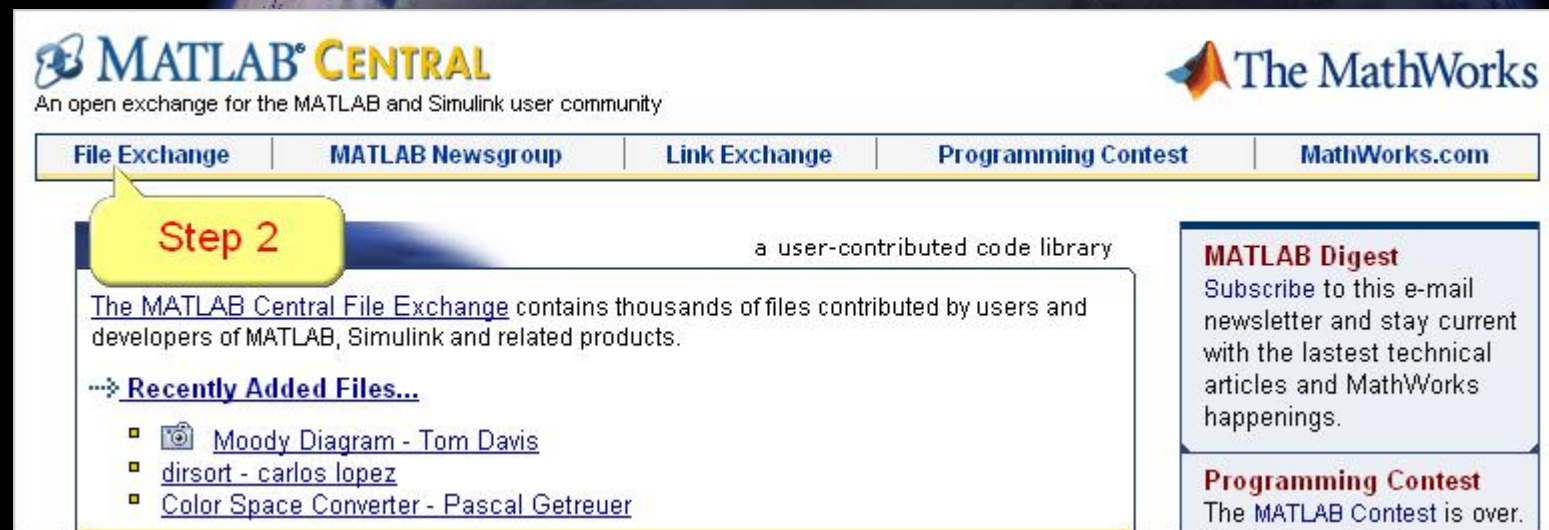
How to get this toolbox :

Step 1:

<http://www.mathworks.com/>



Step 2: Get into “File Exchange”



The screenshot shows the MATLAB Central website interface. At the top, the MATLAB Central logo is on the left, and The MathWorks logo is on the right. Below the logos is a navigation bar with five tabs: File Exchange, MATLAB Newsgroup, Link Exchange, Programming Contest, and MathWorks.com. A yellow callout box labeled "Step 2" points to the File Exchange tab. The main content area features a description of the File Exchange as a user-contributed code library, followed by a section titled "Recently Added Files..." which lists three items: "Moody Diagram - Tom Davis", "dirsort - carlos lopez", and "Color Space Converter - Pascal Getreuer". On the right side, there are two boxes: "MATLAB Digest" with a subscription link and "Programming Contest" with a notice that the contest is over.

MATLAB[®] CENTRAL
An open exchange for the MATLAB and Simulink user community

The MathWorks

[File Exchange](#) | [MATLAB Newsgroup](#) | [Link Exchange](#) | [Programming Contest](#) | [MathWorks.com](#)

Step 2

a user-contributed code library

The [MATLAB Central File Exchange](#) contains thousands of files contributed by users and developers of MATLAB, Simulink and related products.

---> **[Recently Added Files...](#)**

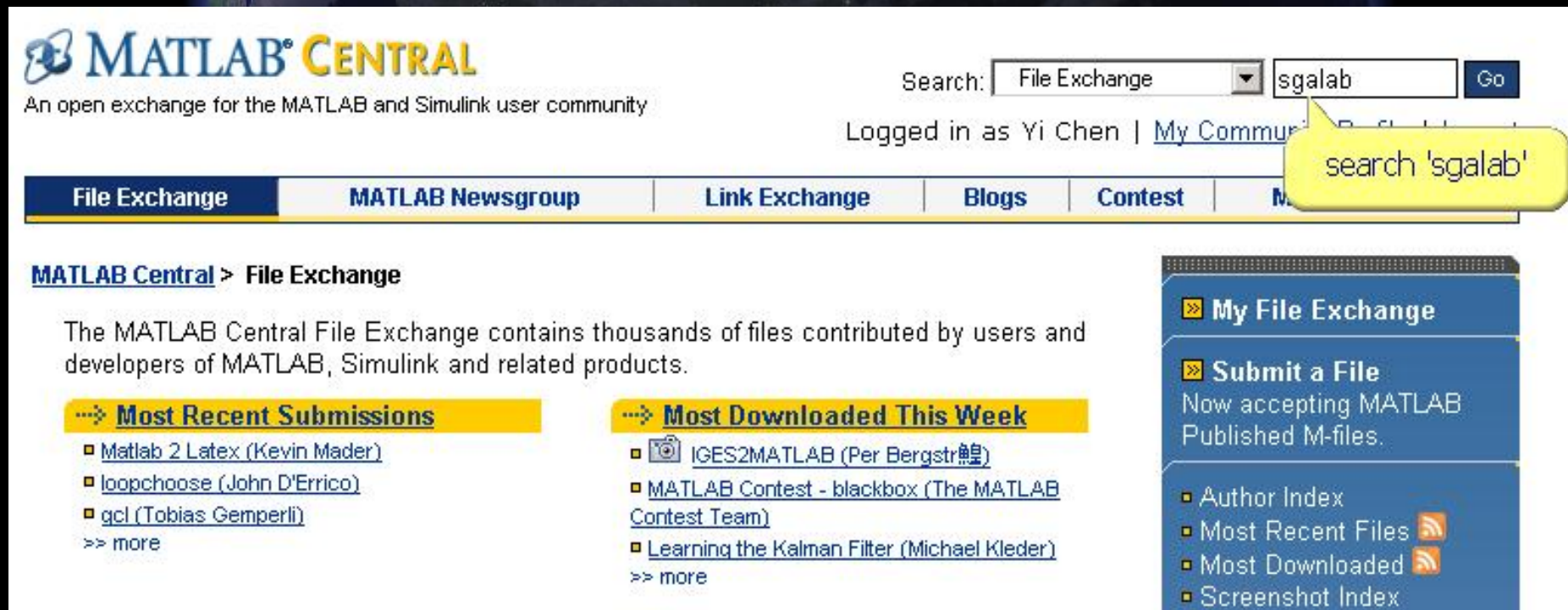
- Moody Diagram - Tom Davis
- dirsort - carlos lopez
- Color Space Converter - Pascal Getreuer

MATLAB Digest
Subscribe to this e-mail newsletter and stay current with the latest technical articles and MathWorks happenings.

Programming Contest
The MATLAB Contest is over.

Genetic Algorithms tools for Matlab(IV) – SGALAB

Step 3:
Search 'SGALAB'



The image shows a screenshot of the MATLAB Central website. At the top, the MATLAB Central logo is displayed with the tagline "An open exchange for the MATLAB and Simulink user community". A search bar is located in the top right, with "File Exchange" selected in the dropdown menu and "sgalab" entered in the search field. A yellow callout bubble points to the search bar with the text "search 'sgalab'". Below the search bar, the user is logged in as "Yi Chen" with links for "My Community" and "My Profile". A navigation bar contains links for "File Exchange", "MATLAB Newsgroup", "Link Exchange", "Blogs", "Contest", and "MATLAB Central". The main content area is titled "MATLAB Central > File Exchange" and contains a description: "The MATLAB Central File Exchange contains thousands of files contributed by users and developers of MATLAB, Simulink and related products." Below this, there are two sections: "Most Recent Submissions" and "Most Downloaded This Week". The "Most Recent Submissions" section lists three items: "Matlab 2 Latex (Kevin Mader)", "loopchoose (John D'Errico)", and "gcl (Tobias Gemperli)", followed by a link to "more". The "Most Downloaded This Week" section lists three items: "IGES2MATLAB (Per Bergström)", "MATLAB Contest - blackbox (The MATLAB Contest Team)", and "Learning the Kalman Filter (Michael Kleder)", followed by a link to "more". On the right side, there is a sidebar with a "My File Exchange" section, a "Submit a File" section with the text "Now accepting MATLAB Published M-files.", and a list of links: "Author Index", "Most Recent Files", "Most Downloaded", and "Screenshot Index".

MATLAB CENTRAL
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Search: File Exchange

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MATLAB Central > File Exchange

The MATLAB Central File Exchange contains thousands of files contributed by users and developers of MATLAB, Simulink and related products.

Most Recent Submissions

- Matlab 2 Latex (Kevin Mader)
- loopchoose (John D'Errico)
- gcl (Tobias Gemperli)

>> more

Most Downloaded This Week

- IGES2MATLAB (Per Bergström)
- MATLAB Contest - blackbox (The MATLAB Contest Team)
- Learning the Kalman Filter (Michael Kleder)

>> more

My File Exchange

Submit a File
Now accepting MATLAB Published M-files.

- Author Index
- Most Recent Files
- Most Downloaded
- Screenshot Index

Search Results for "sgalab"

 Click on column heading to sort by column

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(5=best)

135

 [SGALAB1003 Beta5 \(Inc. NSGA-II\)](#)
Genetic Algorithms Toolbox
Author: [Yi Chen](#)
Category: Medical

2004-09-15

★★★★★
4.45
[27 reviews](#)

80

 [Semi-active Control of Skyhook for 1/4 Suspension System](#)
Semi-active control of skyhook
Author: [Yi Chen](#)
Category: Automotive

2006-05-19

★★★★★
4.67
[3 reviews](#)

» My File Exchange

» Submit a File

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SGALAB1003 Beta5 (Inc. NSGA-II)

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Author:

[Yi Chen](#)

Summary:

Genetic Algorithms Toolbox

MATLAB Release: R14SP2

Required Products: Communications Toolbox

Description:

what's new for SGALAB1003Beta5

1) NSGA-II (SGALAB_demo_MO_NSGAII.m)

2) Pareto front plot & show all Multi-Object Methods in SGALAB (SGALAB_demo_MO_showall.m)

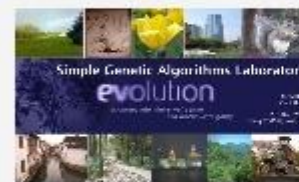
3) documents updated

4) plot bugs fixed for VEGA,MOGA,NSGA,NPGA

Run 'SGALAB_demo_*.m' to see the demos, and to modify the 'SGALAB_demo_*.m' as your main function for your applications

%%%%%%%%%

SGALAB 1003 Beta 4&4 0.0.1



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SGA_encoding

SGA__encoding_binary

SGA__encoding_binary_manually

SGA__encoding_binary_uniformly

SGA__encoding_int_uniformly

SGA__encoding_real_uniformly

SGA__encoding_permutation

SGA__encoding_messy

SGA__encoding_gray

SGA__encoding_DNA

SGA__encoding_DNA_direct

SGA__encoding_DNA_indirect

Simple Genetic Algorithm Laboratory Toolbox

Wizard Control

Step1 : Initialization

Step2 : Cost

Step3 : Selection

Step4 : Crossover

Step5 : Mutation

Step6 : Solve

Step7 : Analysis

Step8 : Export

Status Options

Current Parameters

Time Cost

Process Percent

SGALAB Logfile

Status Off

Status Report

SGALAB1003 GUI get ready ...

Step 1: Initialization

Encoding Method :
Initial Method :

Variables Lower Limit :
Variables Upper Limit :

..\\NPUT_min_confines.txt
..\\NPUT_max_confines.txt

Browse..
Browse..

☐ Multiple objectives

☐ Goal Programming

☐ Non Pareto approaches

☐ Pareto based approaches

Step 1 : Initialization

Quick Reference>>

SGALAB :
Simple Genetic Algorithms Laboratory Toolbo

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Simple Genetic Algorithm Laboratory Toolbox

Wizard Control

- ☐ Step1 : Initialization
- ☒ Step2 : Cost
- ☐ Step3 : Selection
- ☐ Step4 : Crossover
- ☐ Step5 : Mutation
- ☐ Step6 : Solve
- ☐ Step7 : Analysis
- ☐ Step8 : Export

Status Options

- ☐ Current Parameters
- ☐ Time Cost
- ☐ Process Percent
- ☐ SGALAB Logfile
- ☐ Status Off

Status Report

SGALAB1003 GUI get ready ...

Step2 : Cost Evaluation

Fitness Design

☐ Fitness Function

Fitness Scaling :

☐ Fitness System

INPUT File Full Path :

OUTPUT File Full Path :

Constraint Design

☐ Subject to ..

☐ Equal Constraint Functions =

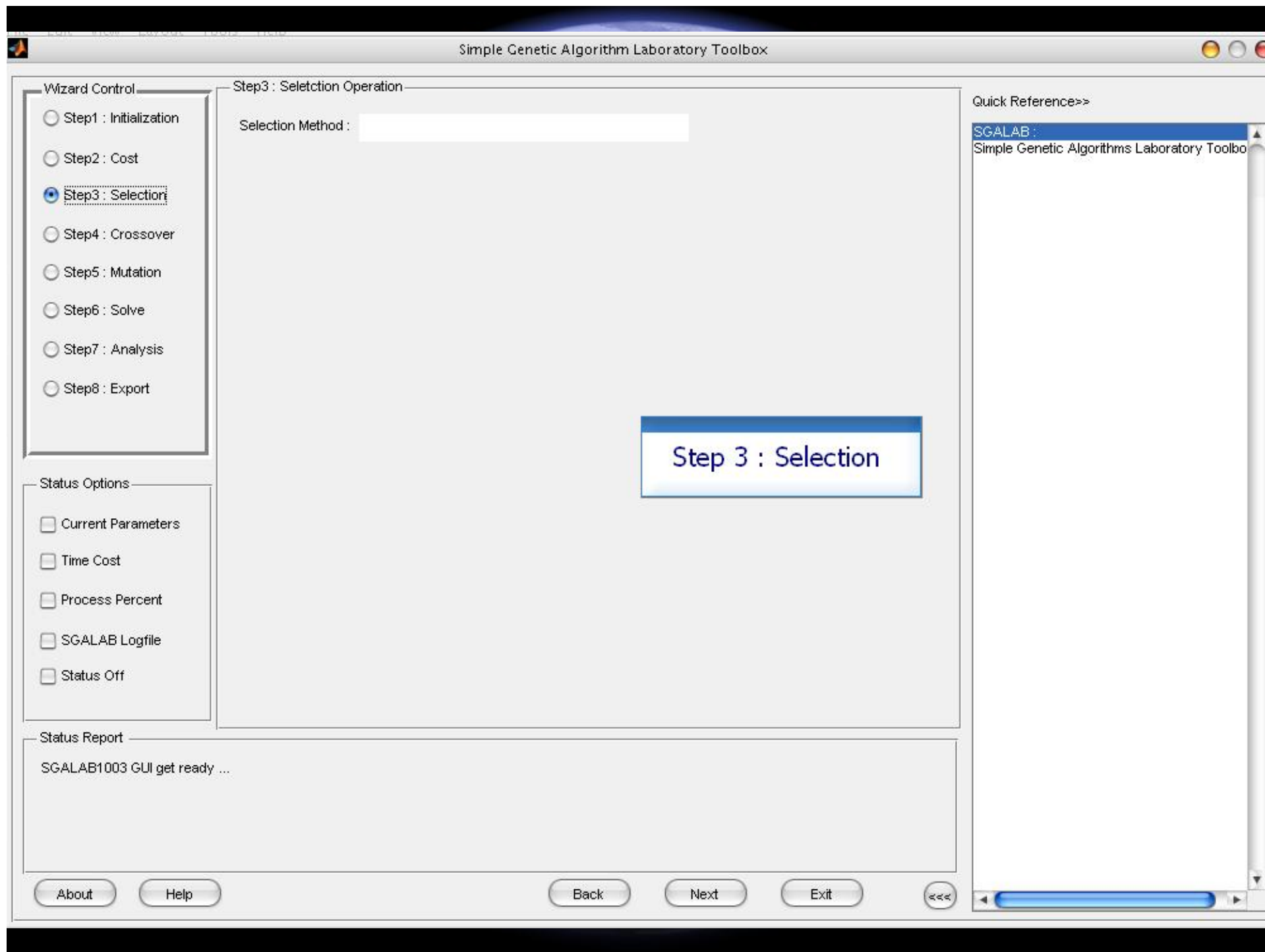
☐ Un-Equal Constraint Functions

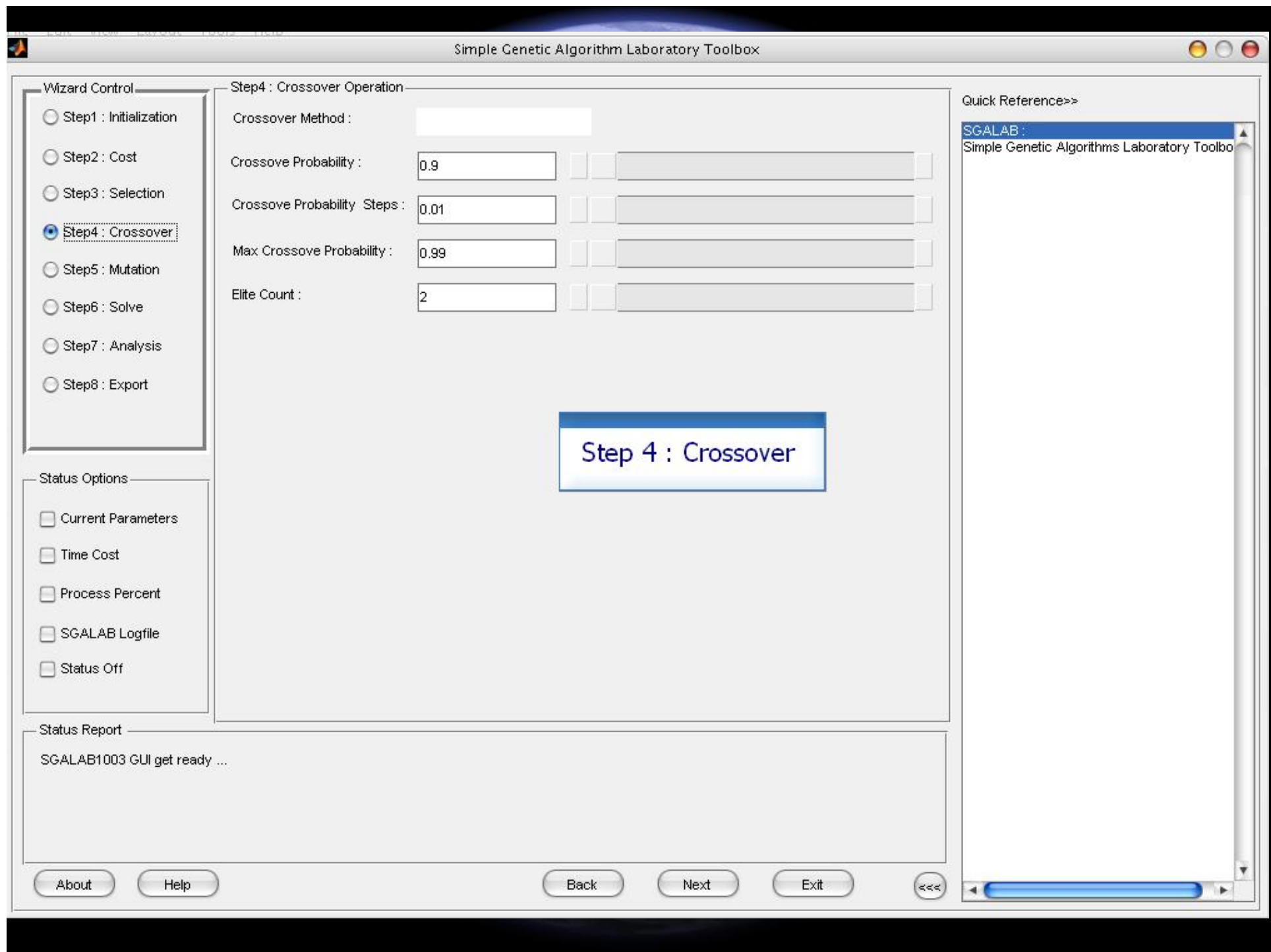
Step 2: Cost

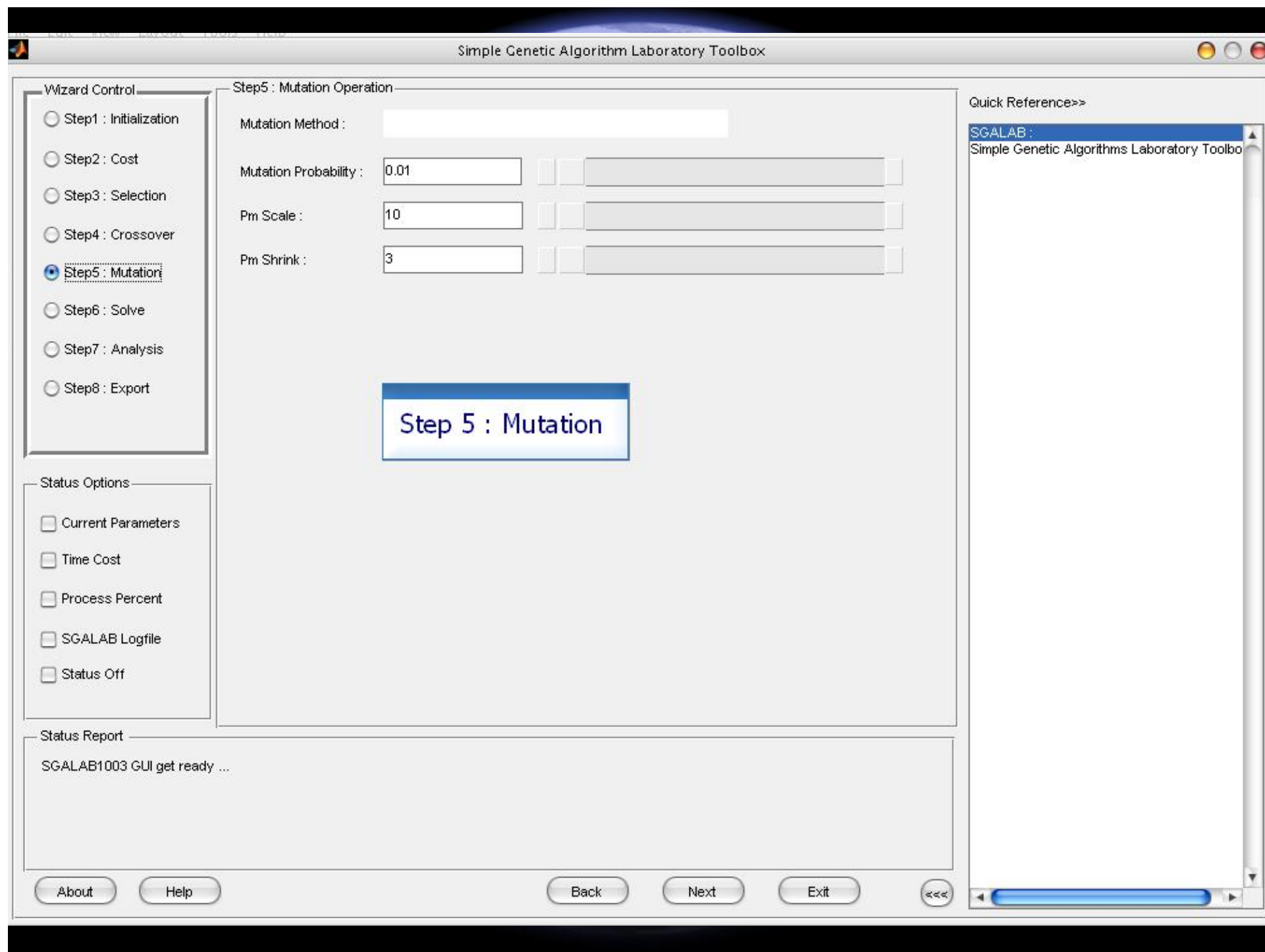
Quick Reference>>

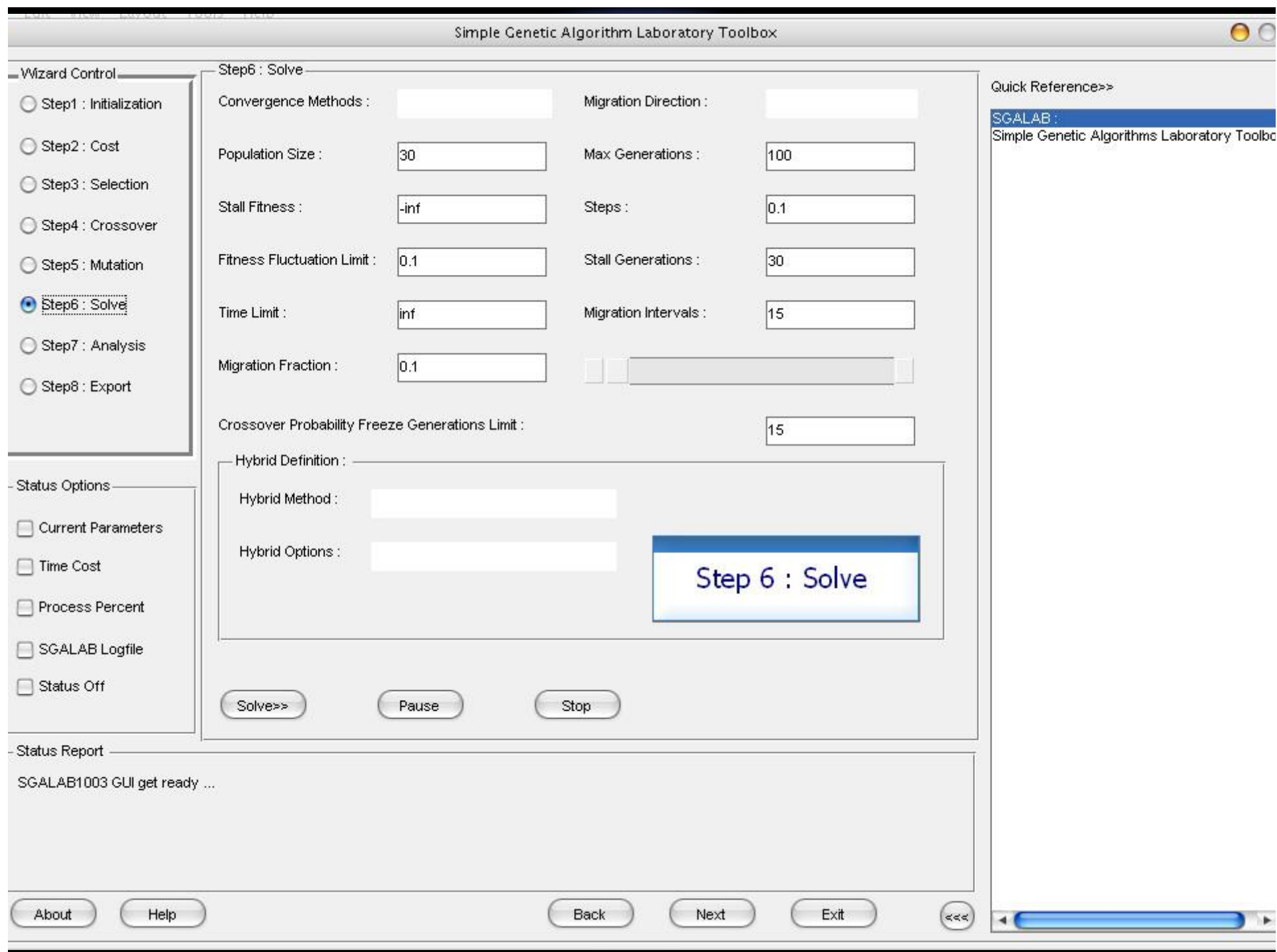
SGALAB :
Simple Genetic Algorithms Laboratory Toolbo

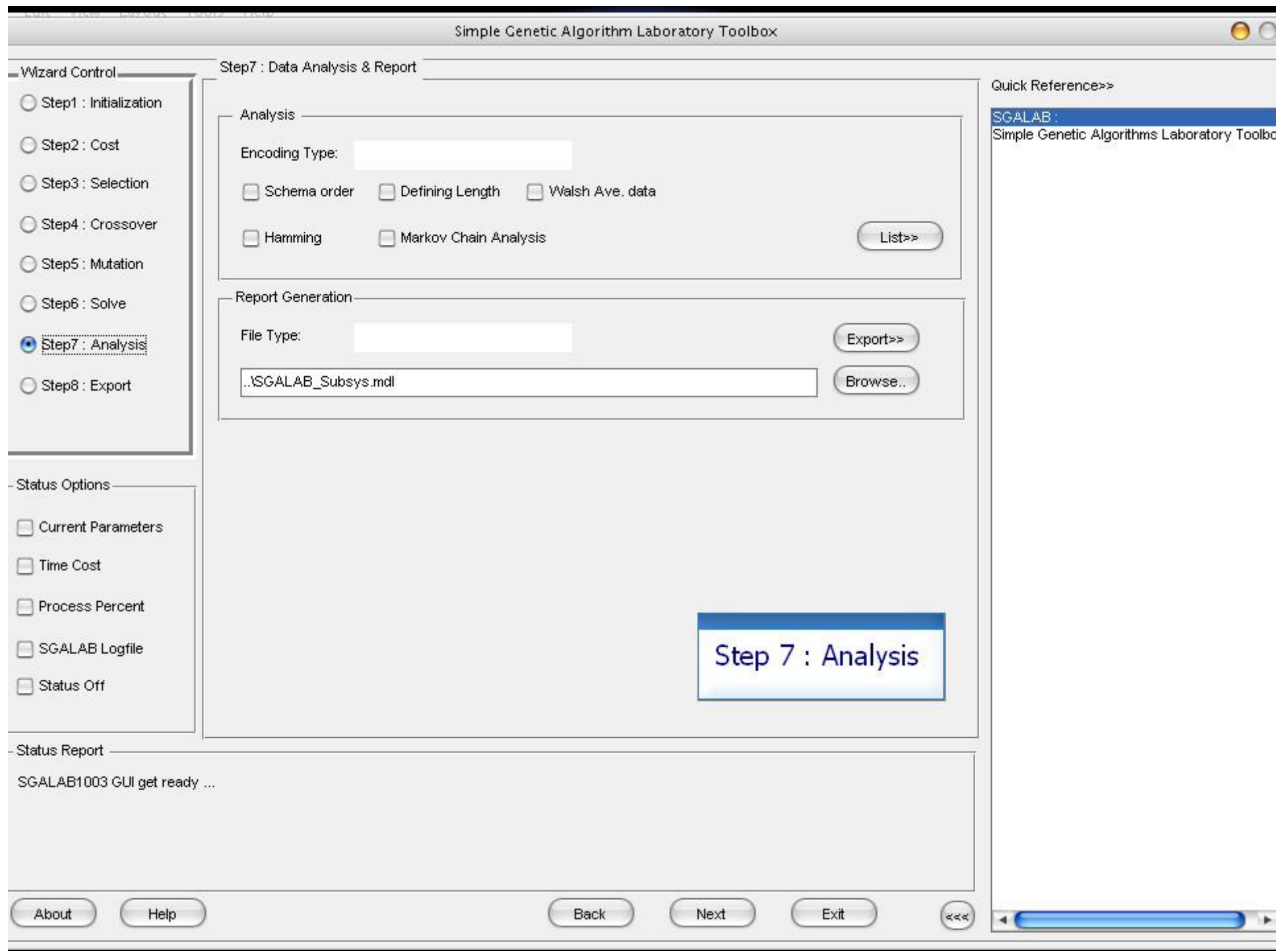
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Wizard Control

- ☐ Step1 : Initialization
☐ Step2 : Cost
☐ Step3 : Selection
☐ Step4 : Crossover
☐ Step5 : Mutation
☐ Step6 : Solve
☐ Step7 : Analysis
☐ Step8 : Export

Status Options

- ☐ Current Parameters
☐ Time Cost
☐ Process Percent
☐ SGALAB Logfile
☐ Status Off

Status Report

SGALAB1003 GUI get ready ...

Step8 : Data Export & Plots

Result List

☐ What to List ?

Result Plots

Generation ~ Fitness

☐ Min. Fitness X Label Generations
☐ Max. Fitness Y Label Fitness Value
☐ Mean Fitness
Plot >> Title Generation vs.Fitness

Generation~Variables~Chromosome

☐ Min. Fitness X Label Generations
☐ Max. Fitness Y Label Variables
☐ Mean Fitness Z Label Chromosome
Plot >> Title Evolution History

Data Export

☐ What to Export ?

Export File Type :

..\OUTPUT_best_result_space.txt

Export>>

Browse..

SGALAB Export to Simulink

Input :

Output :

..\SGALAB_Subsys.mdl

Export>>

Browse..

Quick Reference>>

SGALAB :
Simple Genetic Algorithms Laboratory Toolbox

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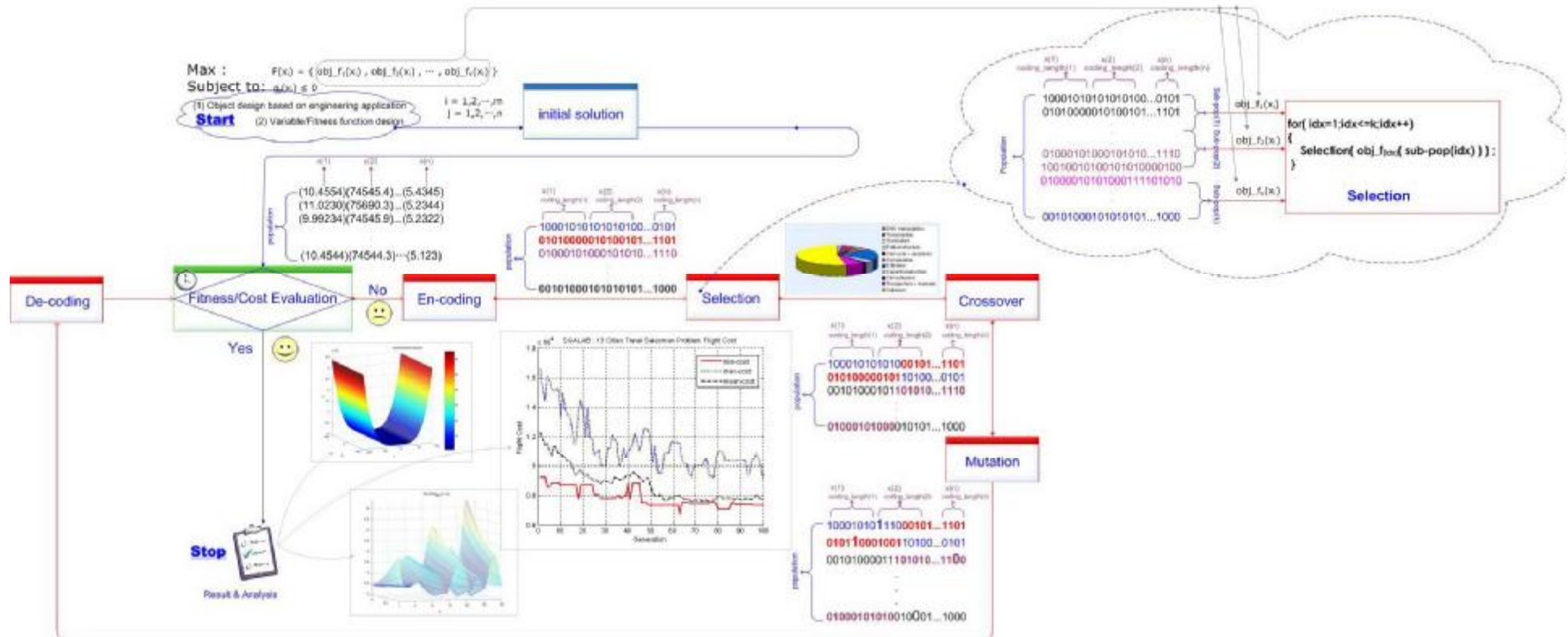
<<<

Step 8: Export

Schaffer - Vector Evaluated Genetic Algorithm(VEGA)

J. David Schaffer, Multiple objective optimization with vector evaluated genetic algorithms, In Genetic Algorithms and their Applications: Proceedings of the First International Conference on Genetic Algorithms, pages 93-103, Hillsdale, NJ, 1985. Lawrence Erlbaum

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26th Nov, 2005



Multiobjective Genetic Algorithm (MOGA)

Fonseca, C. M. & P. J. Fleming: "Genetic algorithms for multiobjective optimization: formulation, discussion and generalization", *Proc. of the 5th Inter. Conf. on GAs*, pp. 416-423, 1993

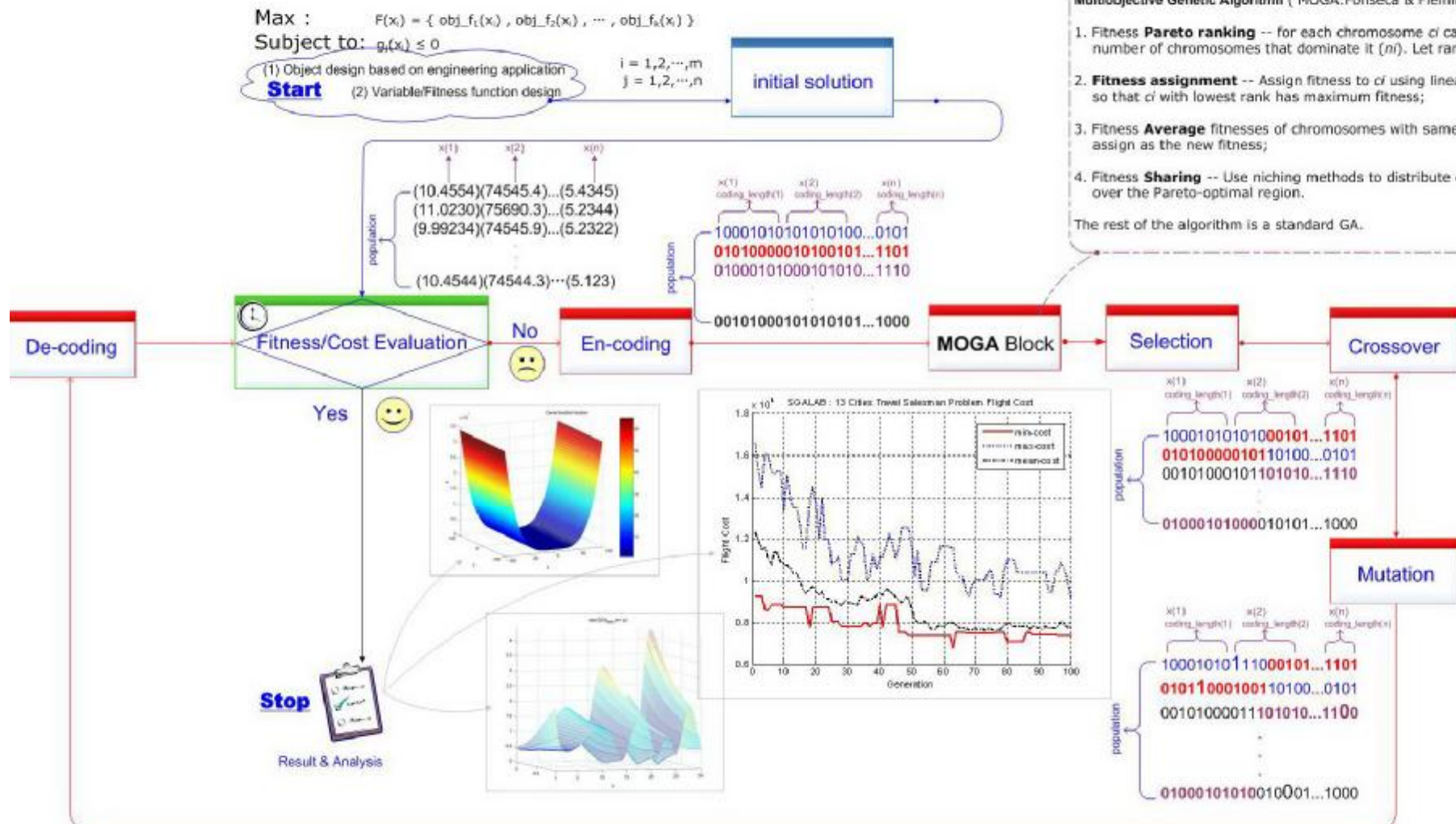
Chen Yi

chenyi2005@gmail.com
28th Nov., 2006

Multiobjective Genetic Algorithm (MOGA: Fonseca & Fleming 1993)

1. Fitness **Pareto ranking** -- for each chromosome ci calculate the number of chromosomes that dominate it (ni). Let rank $ri=1+ni$;
2. **Fitness assignment** -- Assign fitness to ci using linear interpolation so that ci with lowest rank has maximum fitness;
3. Fitness **Average** fitnesses of chromosomes with same rank and assign as the new fitness;
4. Fitness **Sharing** -- Use niching methods to distribute chromosomes over the Pareto-optimal region.

The rest of the algorithm is a standard GA.



Niched Pareto Genetic Algorithm (NPGA)

Horn, J. and N. Nafphtitis (1993). Multiobjective optimization using the niched pareto genetic algorithm. ILLGAL Report 93005, Illinois Genetic Algorithms Laboratory, University of Illinois, Urbana, Champaign.

Chen Yi
chenyi20058@gmail.com
28th Nov., 2005

Max : $F(x) = \{ \text{obj}_1(x), \text{obj}_2(x), \dots, \text{obj}_m(x) \}$

Subject to : $g_i(x) \leq 0$

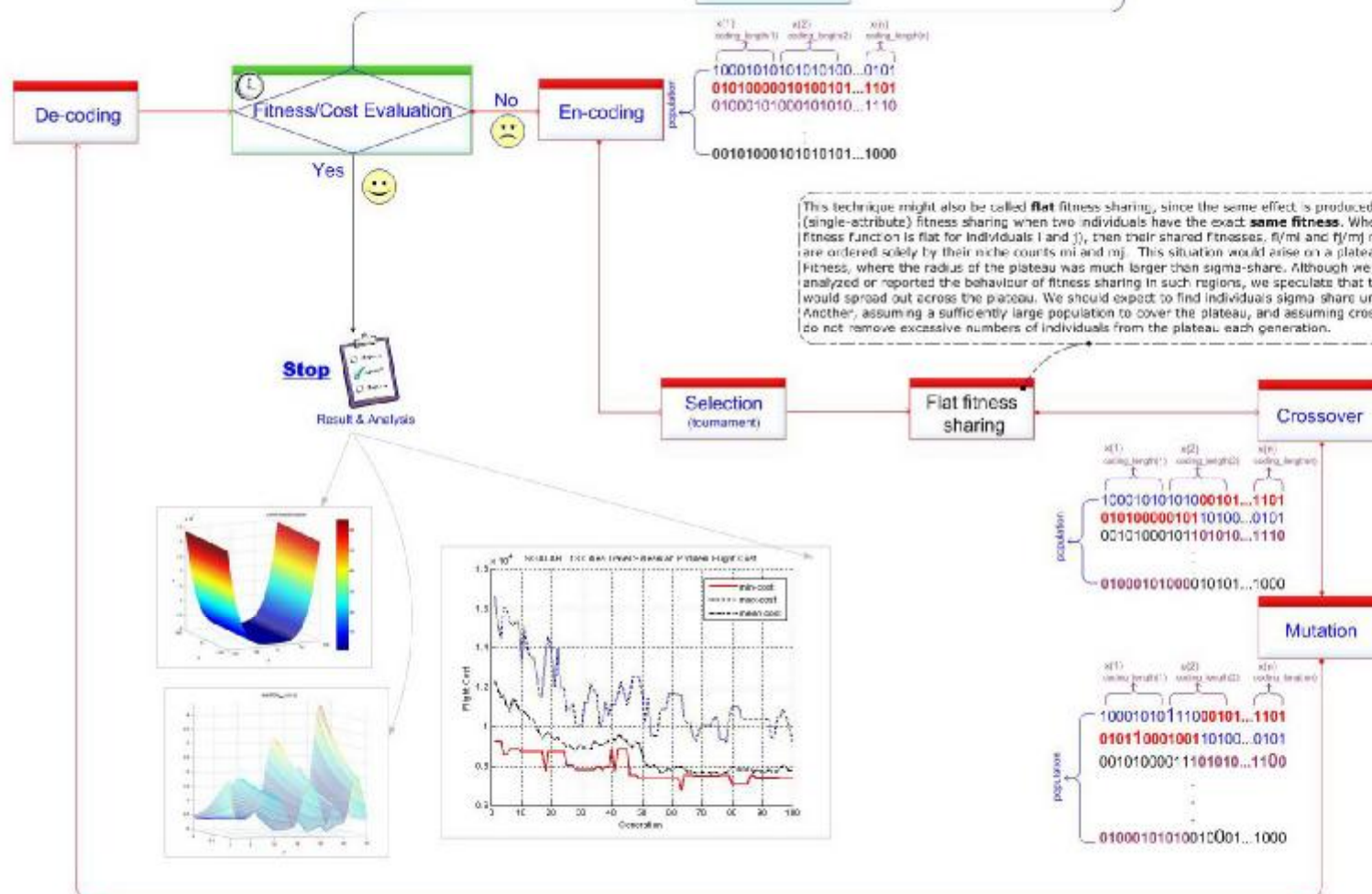
(1) Object design based on engineering application
Start (2) Variable/Fitness function design

$i = 1, 2, \dots, m$
 $j = 1, 2, \dots, n$

initial solution

population
 $x(1) \quad x(2) \quad \dots \quad x(n)$
(10.4554)(74545.4)...(5.4345)
(11.0230)(75690.3)...(5.2344)
(9.99234)(74545.9)...(5.2322)
...
(10.4544)(74544.3)...(5.123)

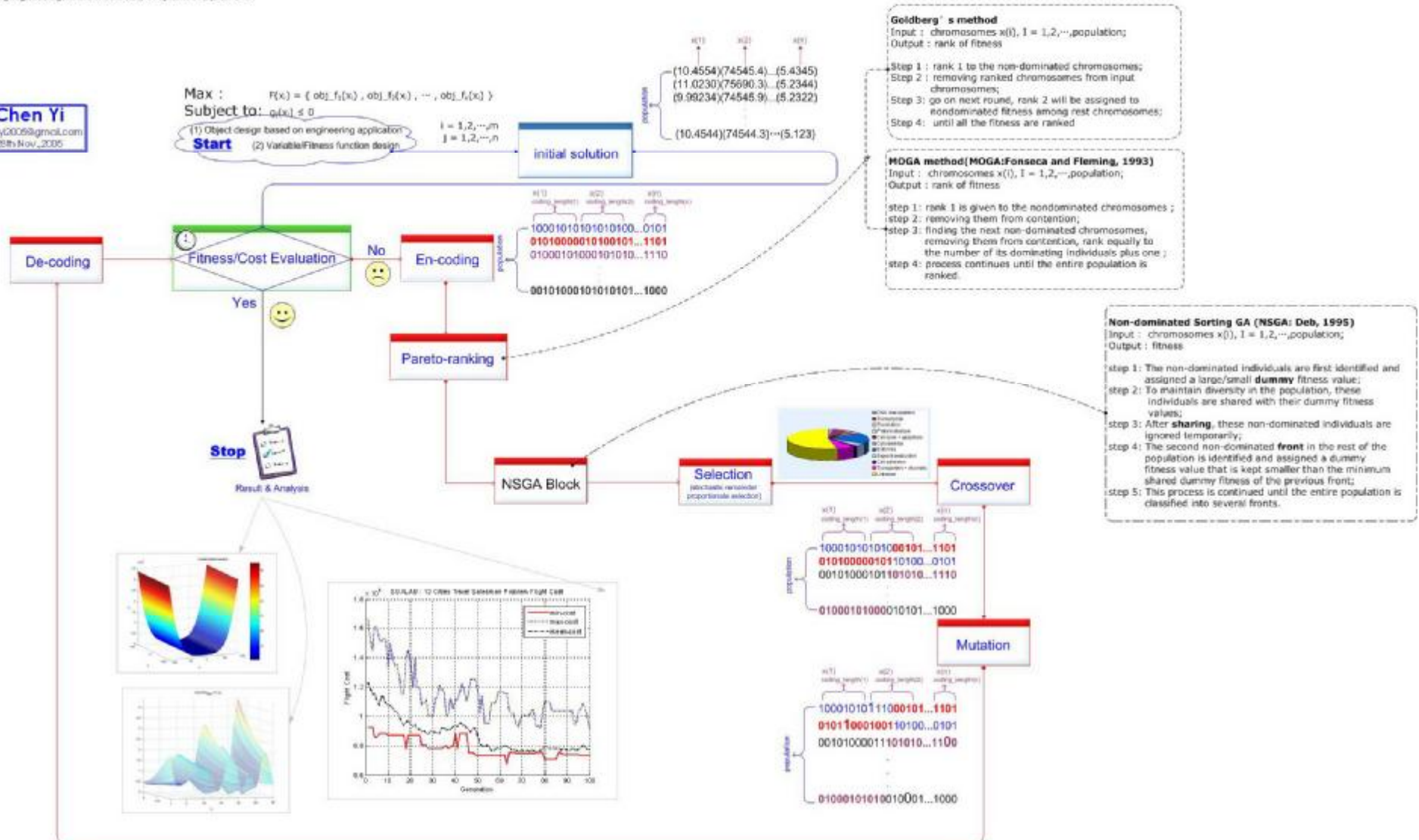
Niche count $n_i = \sum_{j=1}^{\text{population}} \delta(x_i, x_j)$
 $i = 1 - \text{population}$
population
 $x(1) \quad x(2) \quad \dots \quad x(n)$
(10.4554)(74545.4)...(5.4345)
(11.0230)(75690.3)...(5.2344)
(9.99234)(74545.9)...(5.2322)
...
(10.4544)(74544.3)...(5.123)



Non-dominated Sorting Genetic Algorithm(NSGA)

Srinivas, N. and K. Deb (1994) Multiobjective optimization using nondominated sorting in genetic algorithms. *Evolutionary Computation* 2(3):221-248.

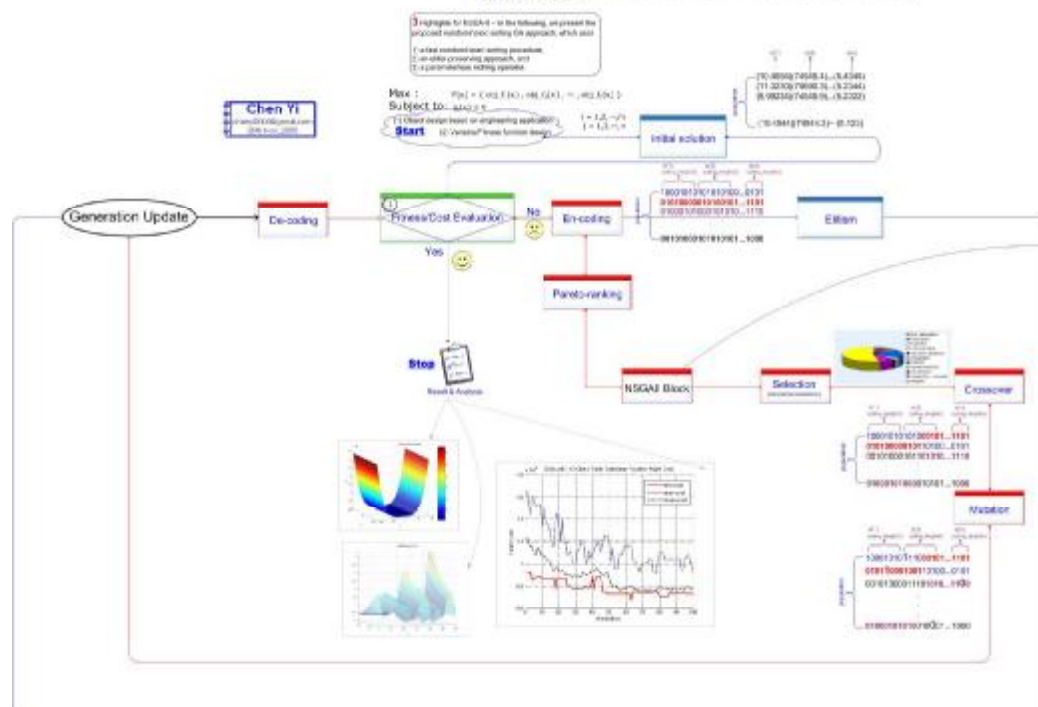
Chen Yi
chany2009@gmail.com
25th Nov., 2005





Non-dominated Sorting Genetic Algorithm-II(NSGA-II)

Deb, K. Multi-objective optimization using Evolutionary Algorithms. John Wiley, 2001.
Deb, K., Pratap, A., Agarwal, S., & Meyarivan, N. Fast and elitist Multiobjective Genetic Algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation, vol. 6, no. 2, pp. 182-197, 2002.



Fast non-dominated sort II

```

{
  Step 1) pareto rank
  Step 2) get fronts
  Step 3) sort population into different groups according to fronts
  Step 4) crowding distance assignment
}

```

NSGAII goes as following steps:

Step 1: Fast non-dominated sort
Step 2: crowding distance assignment
Step 3: crowded-comparison operator

Crowding_distance_assignment()

/* In current Front, there are n fitness functions, n = 1, 2, ..., N */

```

{
  (1) Set initialization distance to 0,
  Dist[2:n-1] = 0,
  Dist[1] = Dist[n] = inf

  (2) MAX & MIN sort objective value, namely,
  MAX & MIN sort( fitness )

  (3) get each distance by following function:

```

```

for i=2 to n-1
  for j=1 to n
    if (j < i)
      Dist[i] = min(Dist[i], (F[j,1] - F[i,1]) / (F[j,n] - F[i,n]))
    if (j > i)
      Dist[i] = min(Dist[i], (F[j,1] - F[i,1]) / (F[j,n] - F[i,n]))
  }
}

```

crowded-comparison operator(ranks, crowded_distances)

```

{
  i < j
  if (rank[i] < rank[j])
  or
  { rank[i] == rank[j] && distance[i] > distance[j] }
  get i;
else
  get j;
}

```

crowded-comparison operator: The crowded-comparison operator ($\leq_c$) prefers the solution which is less crowded than the other. At any time that every individual i in the population has two attributes:

1) nondomination rank(r_{rank});
2) crowding distance(d_{crowd});
We now define a partial order $\leq_c$ as:

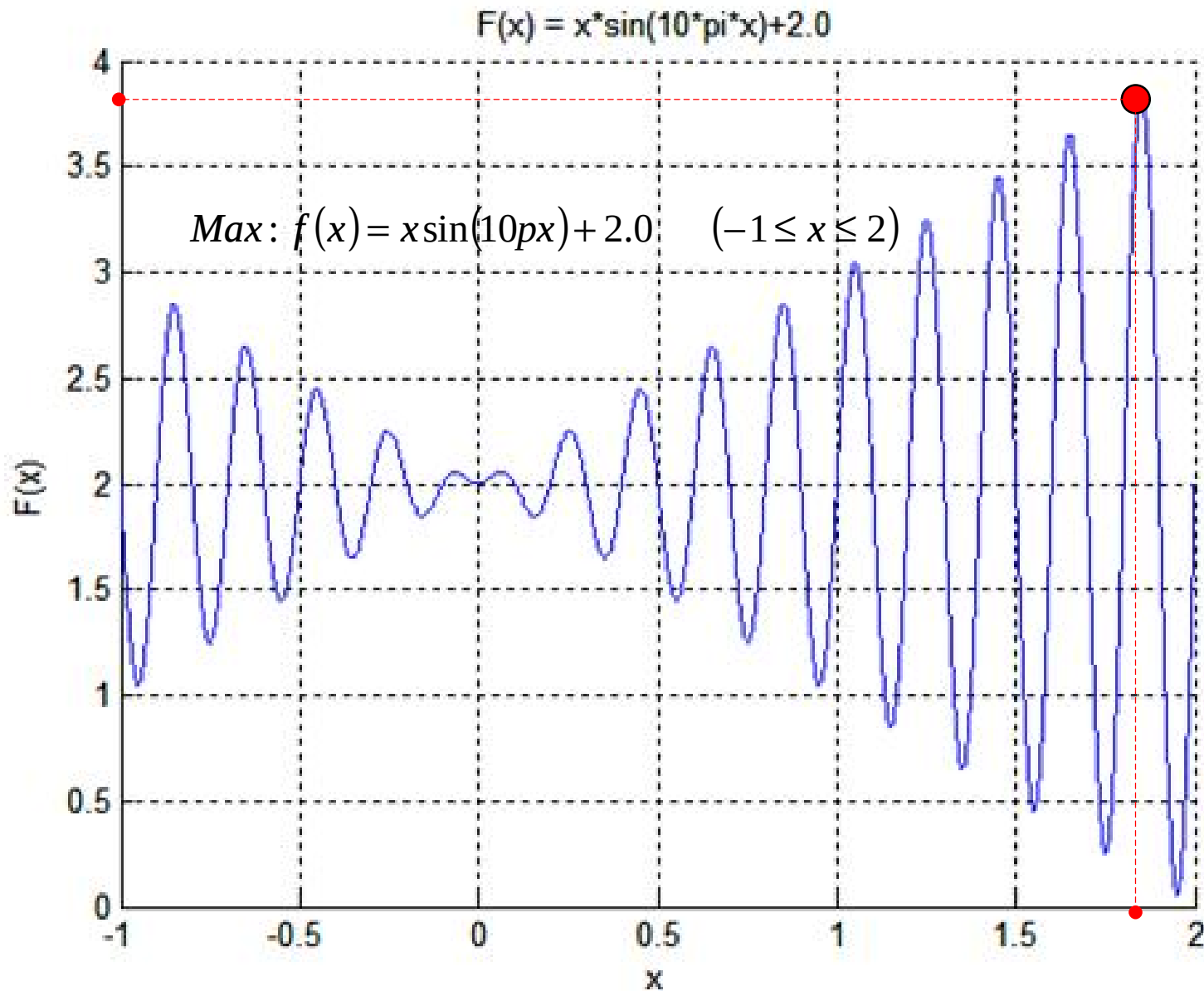
Two solutions i and j are compared by their rank. If $r_{rank}[i] < r_{rank}[j]$, then we prefer i to j . If both solutions belong to the same front, then we prefer the solution that is located in a less crowded region. With these two new attributes – a fast non-dominated sorting procedure, a fast crowding distance estimation procedure, and a simple crowded-comparison operator, we are now ready to describe the NSGA-II algorithm.

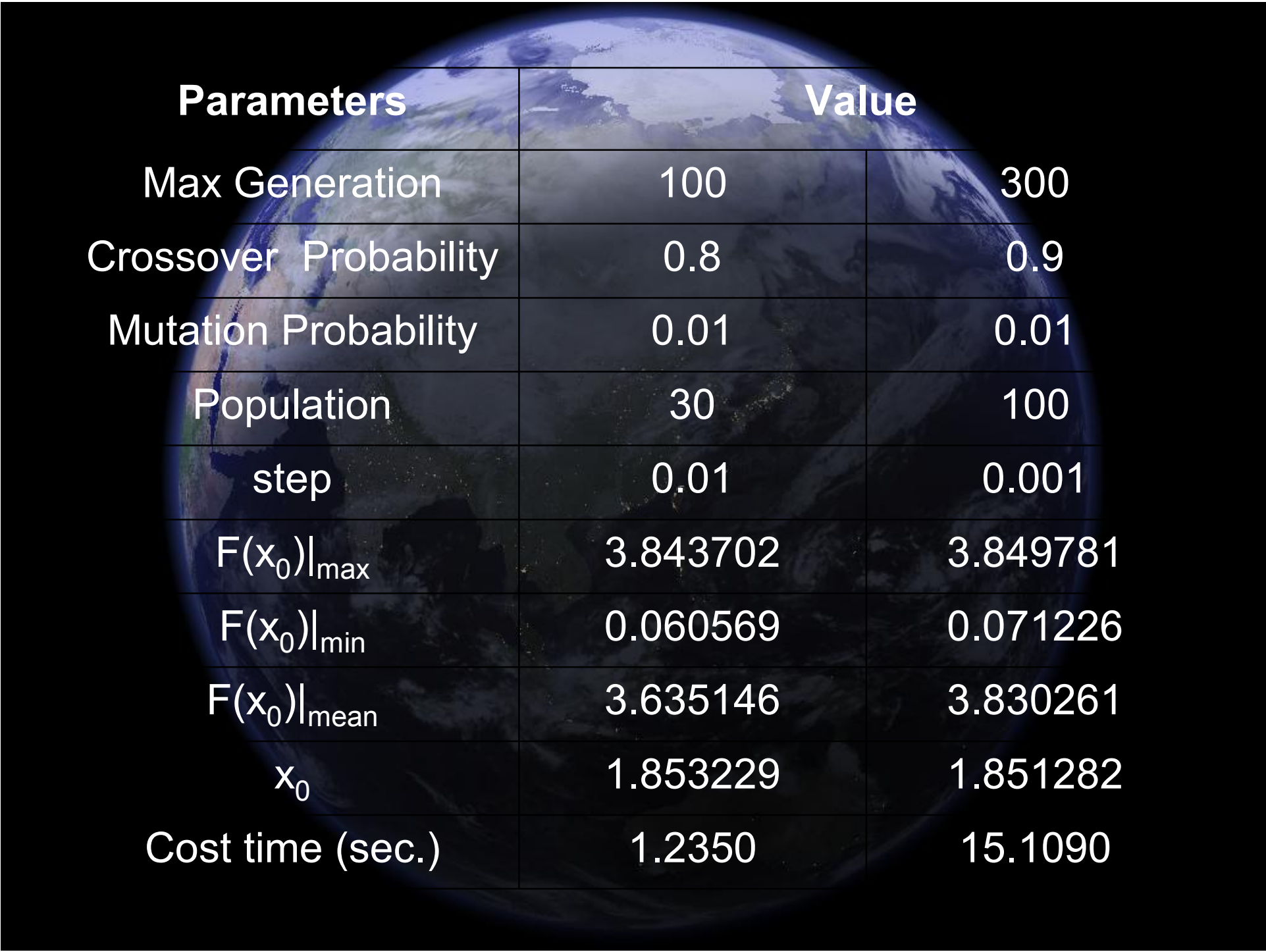
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A Simple Example (using SGALAB1001)

Unconstrained Numerical Optimization (Michalewicz, 1996)

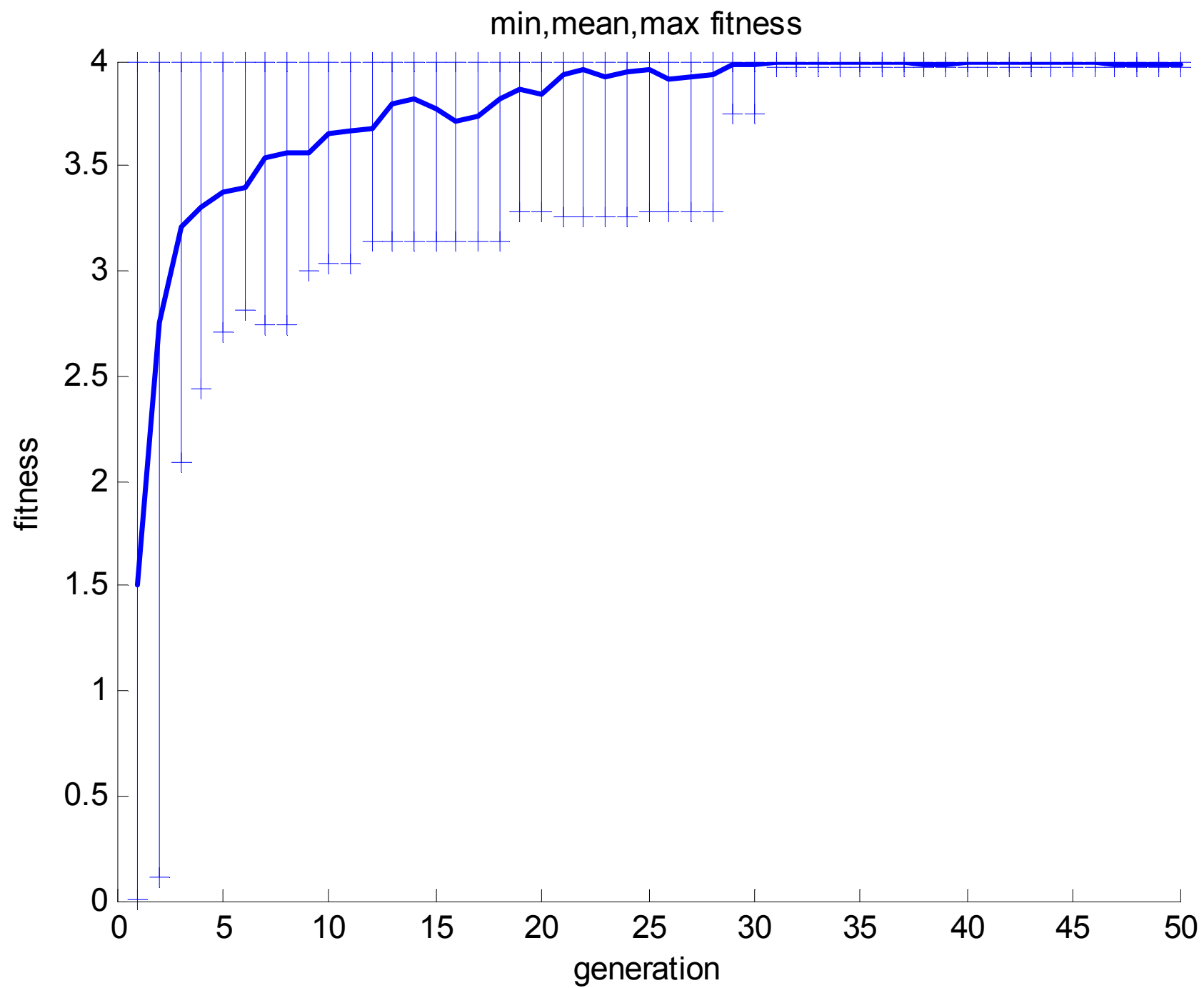




Parameters	Value	
Max Generation	100	300
Crossover Probability	0.8	0.9
Mutation Probability	0.01	0.01
Population	30	100
step	0.01	0.001
$F(x_0) _{\max}$	3.843702	3.849781
$F(x_0) _{\min}$	0.060569	0.071226
$F(x_0) _{\text{mean}}$	3.635146	3.830261
x_0	1.853229	1.851282
Cost time (sec.)	1.2350	15.1090

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TSP 13 Cities Demo for SGALAB1003

The 'SGALAB' will take a world wide show of its new product -SGALAB 1003 , there are 13 cities on the list of its Biz-traveling.
Give 1~13 numbers to each city as following:

Num.	City
1	<i>San Francisco</i>
2	<i>New York</i>
3	<i>Brasilia</i>
4	<i>Madrid</i>
5	<i>London</i>
6	<i>Stockholm</i>
7	<i>Johannesburg</i>
8	<i>Moscow</i>
9	<i>New Delhi</i>
10	Chongqing
11	<i>Shanghai</i>
12	<i>Tokyo</i>
13	<i>Canberra</i>

The start point is the city of Chongqing (10), how can SGALAB company make a plan for its world wide Biz-travel with lowest traveling cost?

Simulation Parameters Setting

Parameters	Value
Population	50
Max Generation	100
City-Number	13
Start city	10
Pc (CX)	0.9
Pm(Insertion)	0.01
TSP_cost_matrix	Appendix1
Run time	7.172000

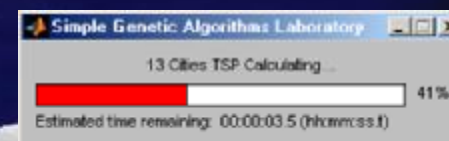
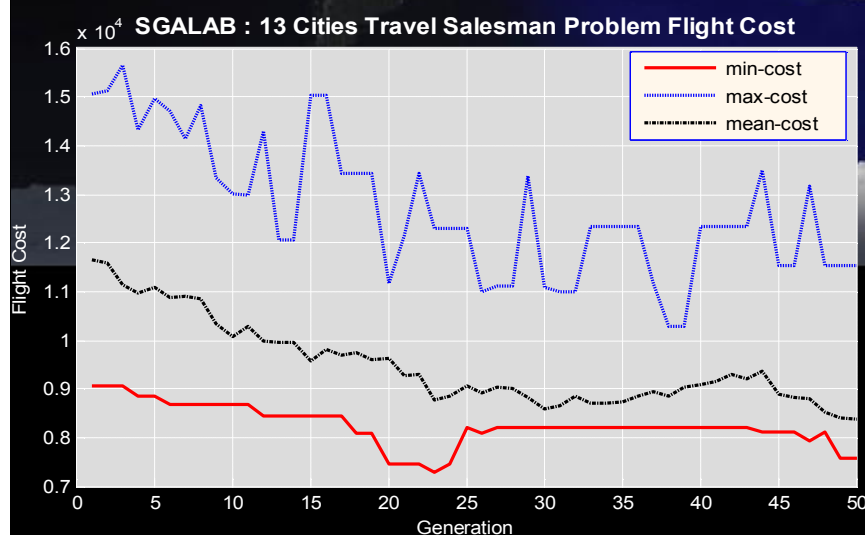
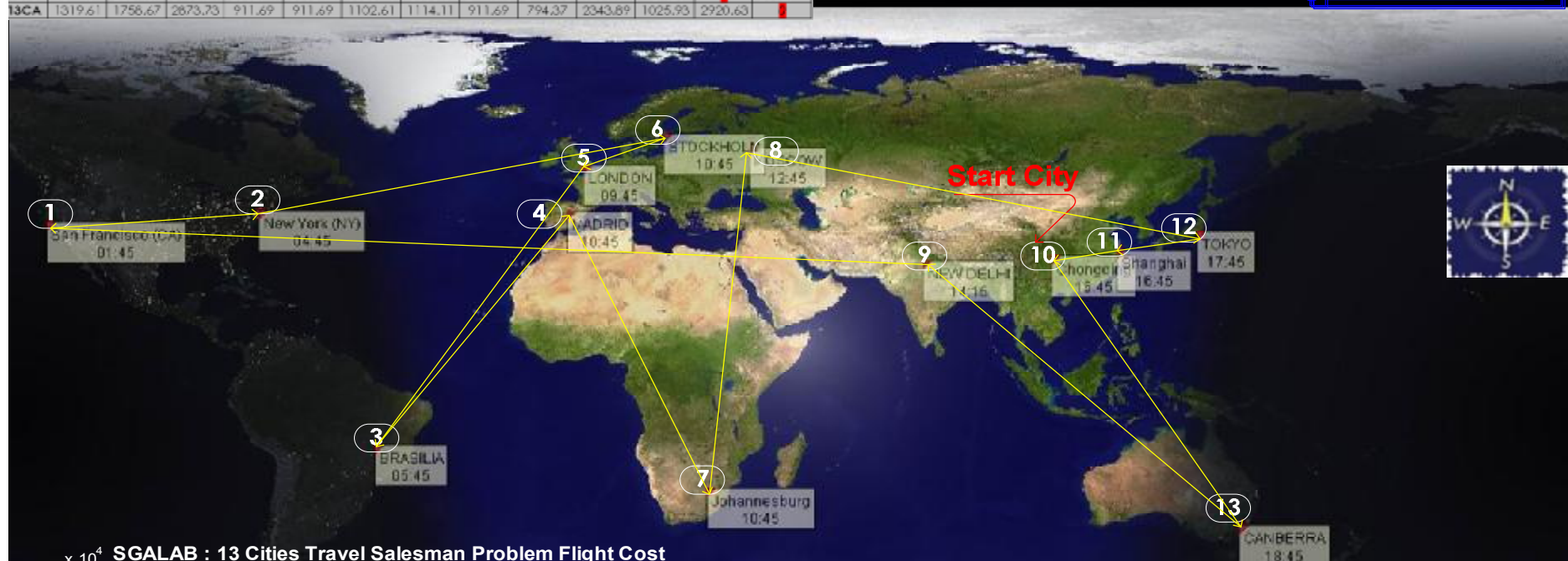
Cost	1SA	2NY	3BR	4MA	5LO	6ST	7JO	8MO	9ND	10CQ	11SH	12TO	13CA
1SA		106.05	1345	845	845	1295	845	607	656.59	525	1019	735.34	
2NY	106.05		916	575	297	381	969	685	604	775.59	644	359	933.34
3BR	1109	999		784	919	1068	1279	1059	1643	2145	1971	1266	1954.68
4MA	382.95	273.71	882.89		44.18	79.78	352.26	239.34	1042.07	942.65	365.76	870.23	2163.71
5LO	564.38	496.88	632.54	47.49		18.26	465.82	155.27	405.54	886.89	548.03	652.16	1296.52
6ST	1374.72	260.9	877.05	54.18	18.06		702.44	124.43	434.84	936.59	634.87	600.09	2864.24
7JO	948.65	816.11	2054.74	610.59	580.81	647.82		610.59	707.39	1496.76	896.53	787.81	2696.92
8MO	509	315	610	291	170	145	433		307	774	525	700	1286.8
9ND	597.23	568.52	1519.73	384.98	372.12	385.9	551.29	385.9		546.69	470.89	493.86	1804.95
10CQ	703.19	836.1	2510.72	927.92	927.92	927.92	1496.76	1074.72	712.86		180.02	918.26	1767.14
11SH	555.79	459.13	1555	490.29	398.71	473.63	146.31	451.88	567.87	180.02		459.13	1753.85
12TO	1647.04	2229.67	3952.34	2352.91	2352.91	3323.96	3006.5	2711.45	2117.62	1266.09	427.26		2582.71
13CA	1319.61	1758.67	2873.73	911.69	911.69	1102.61	1114.11	911.69	794.37	2343.89	1025.93	2920.63	

13 Cities Traveling Salesman Problem

Simple Genetic Algorithms Laboratory (SGALAB) 1.0.0.3 for Matlab 7.x
31st Nov., 2005

Chen Yi

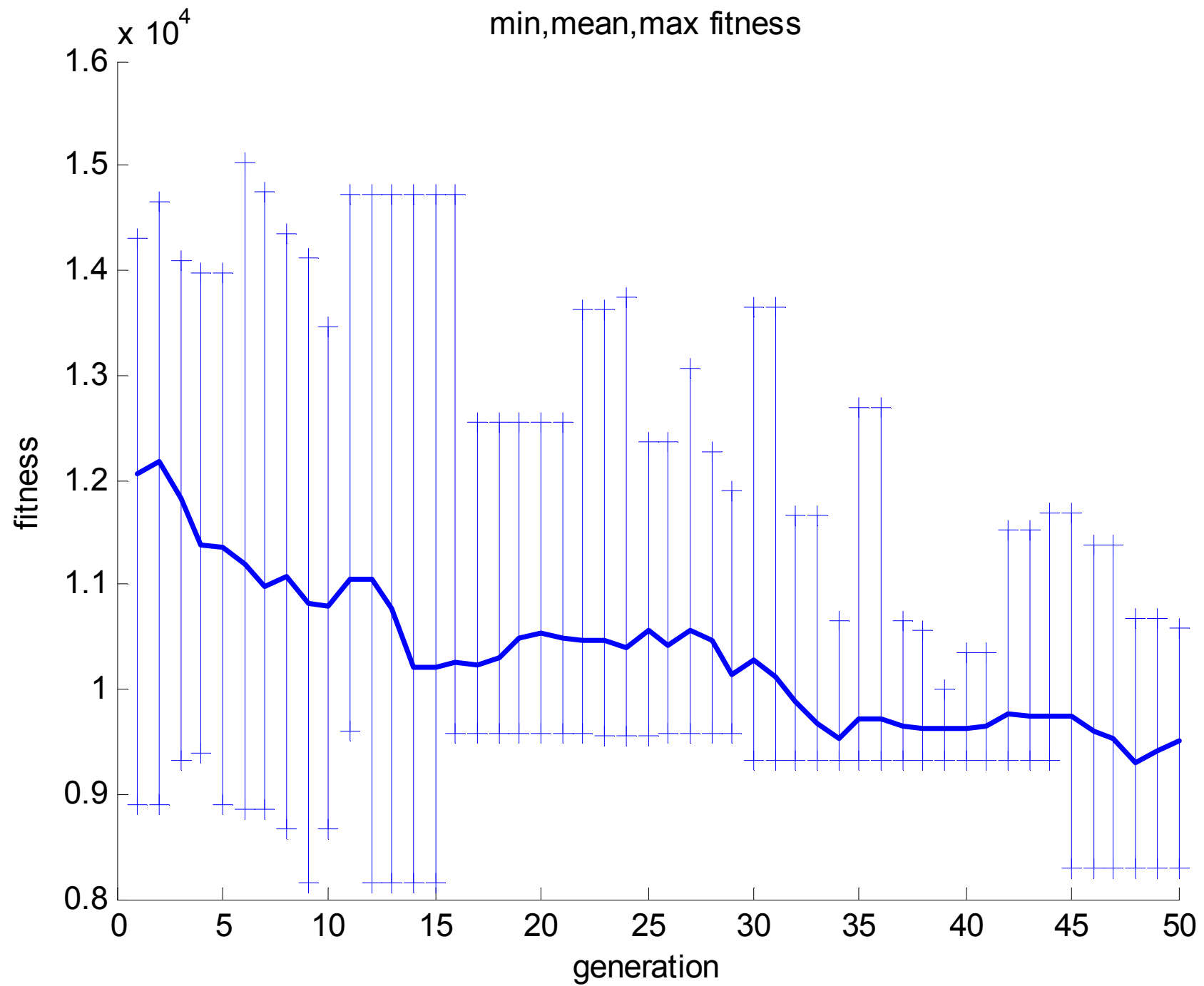
chenyi2005@gmail.com



Best TSP Path :

10 → 13 → 9 → 1 → 2 → 6 → 5 → 3 → 4 → 7 → 8 → 12 → 11 → 10

Lowest Flight Cost : 7.3505e+003



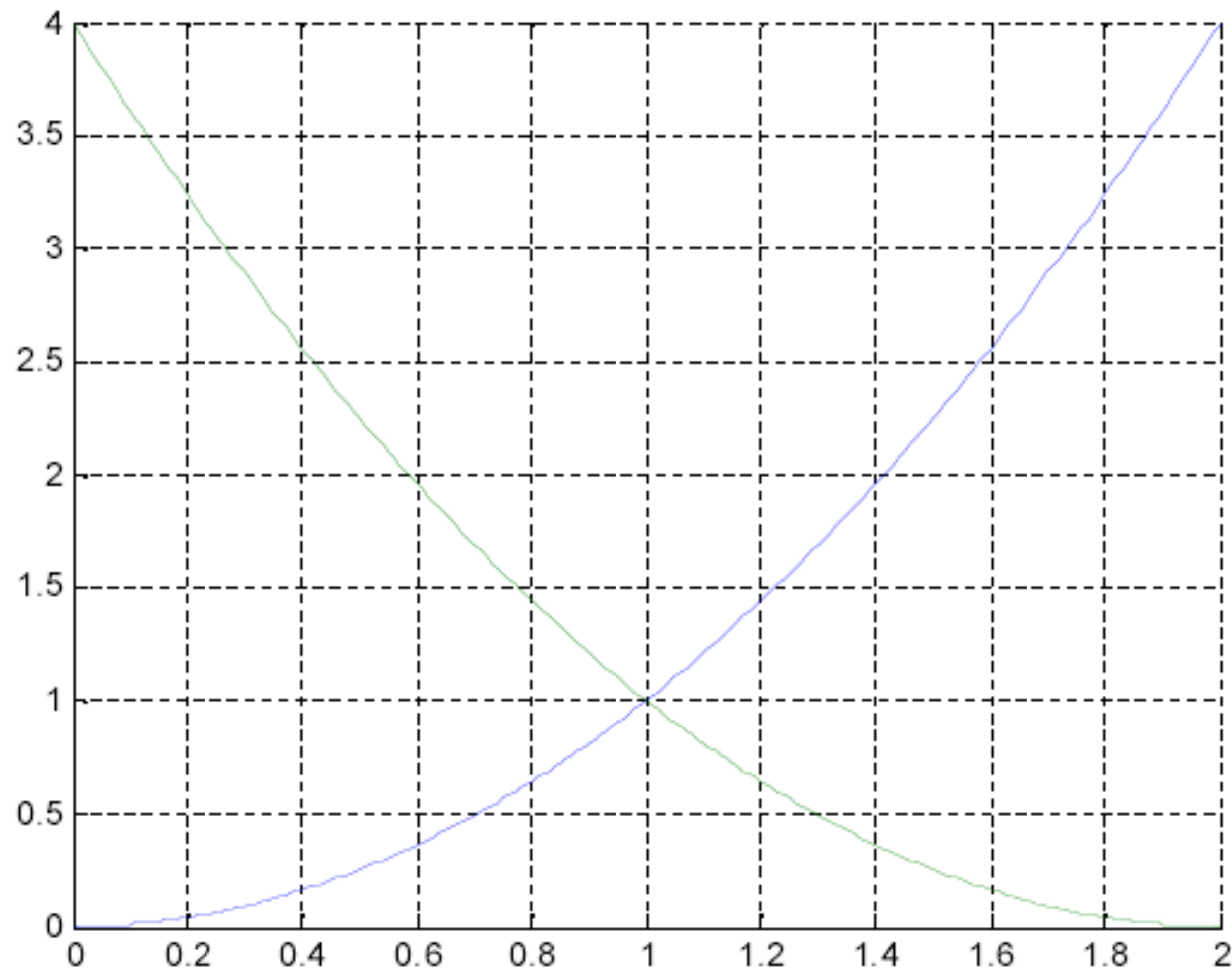
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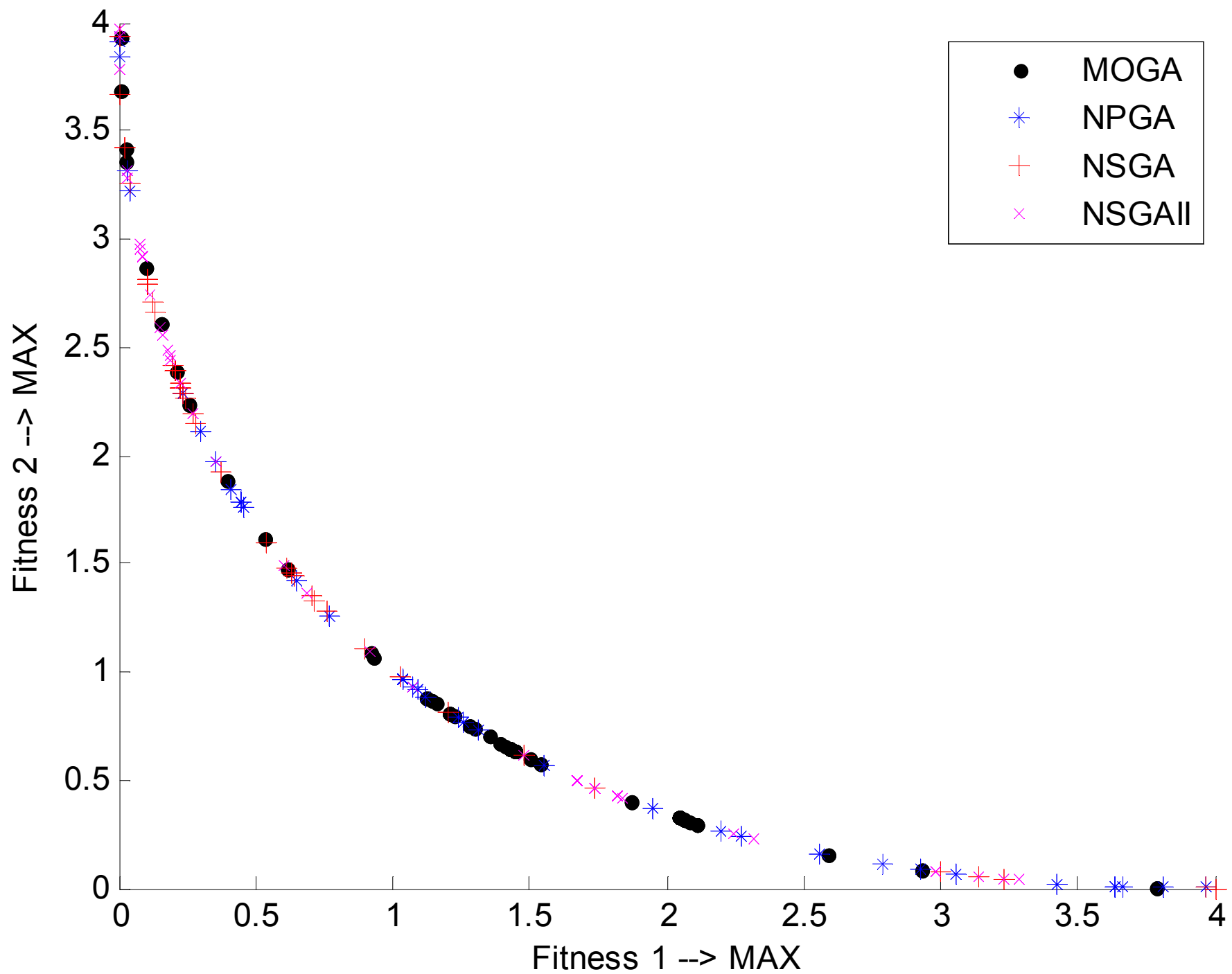
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Demo3 - A multi-objective problem

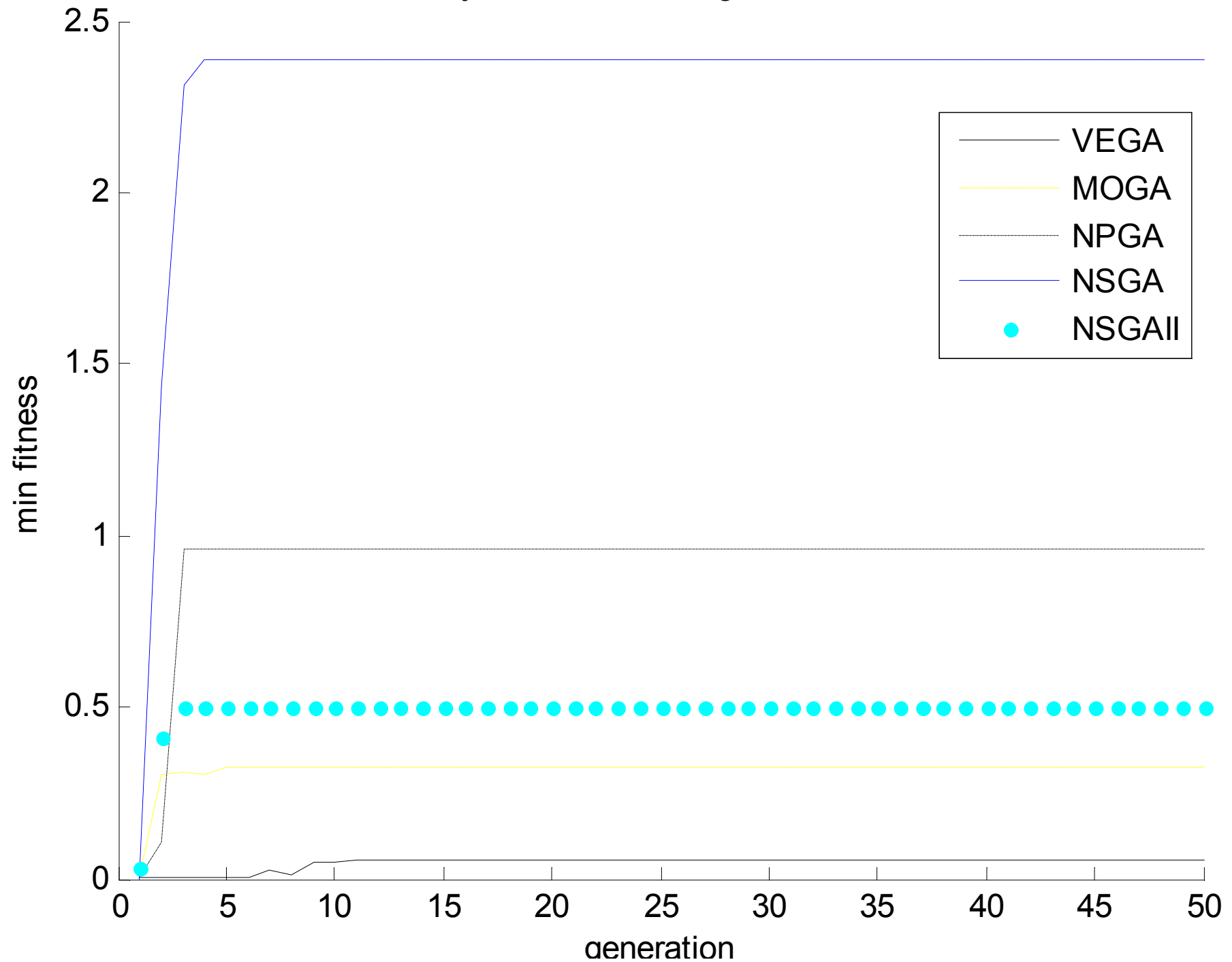
Assume the following 2 fitness ,

$$\begin{cases} f_1(x) = x^2 \\ f_2(x) = (x-2)^2 \end{cases} \quad x \in [0,2] \quad , \text{ so } f_1 = f_2 = 1 \text{ should be the best results}$$





Multi-Objective Genetic Algorithms Methods

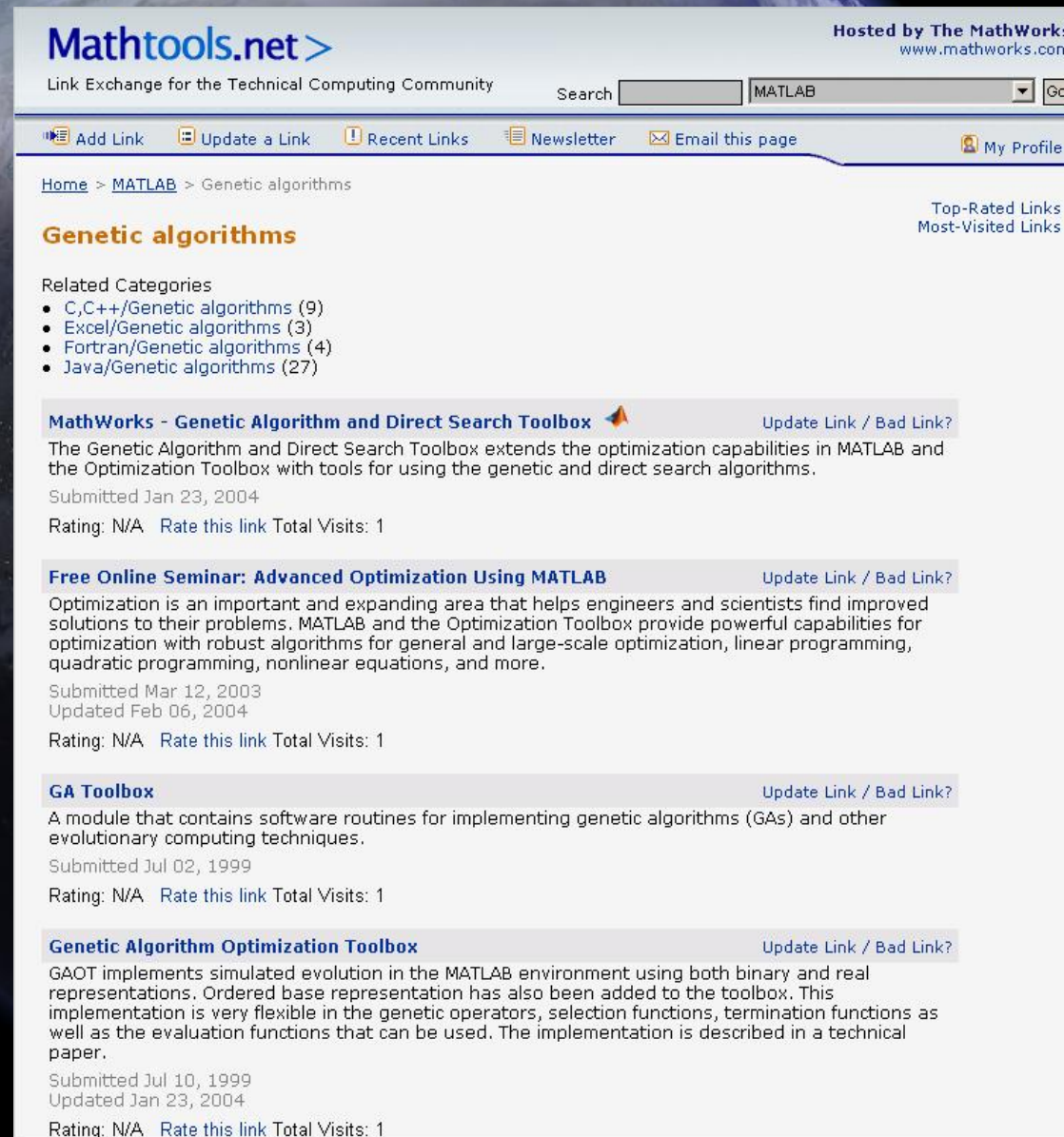


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Some Links where have GAs discussion :

(1) http://www.mathtools.net/MATLAB/Genetic_algorithms/



The screenshot shows the Mathtools.net website interface. At the top, there's a header with the site name "Mathtools.net" and a tagline "Link Exchange for the Technical Computing Community". A search bar contains the word "MATLAB". Below the header, there are navigation links: "Add Link", "Update a Link", "Recent Links", "Newsletter", "Email this page", and "My Profile". The main content area is titled "Genetic algorithms" and includes a list of "Related Categories" such as "C,C++/Genetic algorithms (9)", "Excel/Genetic algorithms (3)", "Fortran/Genetic algorithms (4)", and "Java/Genetic algorithms (27)". There are three main link entries, each with a title, description, submission date, and rating. The first entry is "MathWorks - Genetic Algorithm and Direct Search Toolbox", the second is "Free Online Seminar: Advanced Optimization Using MATLAB", and the third is "GA Toolbox". Each entry has a "Rate this link" button and a "Total Visits" count.

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[Home](#) > [MATLAB](#) > Genetic algorithms

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Related Categories

- C,C++/Genetic algorithms (9)
- Excel/Genetic algorithms (3)
- Fortran/Genetic algorithms (4)
- Java/Genetic algorithms (27)

MathWorks - Genetic Algorithm and Direct Search Toolbox [Update Link / Bad Link?](#)

The Genetic Algorithm and Direct Search Toolbox extends the optimization capabilities in MATLAB and the Optimization Toolbox with tools for using the genetic and direct search algorithms.

Submitted Jan 23, 2004

Rating: N/A [Rate this link](#) Total Visits: 1

Free Online Seminar: Advanced Optimization Using MATLAB [Update Link / Bad Link?](#)

Optimization is an important and expanding area that helps engineers and scientists find improved solutions to their problems. MATLAB and the Optimization Toolbox provide powerful capabilities for optimization with robust algorithms for general and large-scale optimization, linear programming, quadratic programming, nonlinear equations, and more.

Submitted Mar 12, 2003
Updated Feb 06, 2004

Rating: N/A [Rate this link](#) Total Visits: 1

GA Toolbox [Update Link / Bad Link?](#)

A module that contains software routines for implementing genetic algorithms (GAs) and other evolutionary computing techniques.

Submitted Jul 02, 1999

Rating: N/A [Rate this link](#) Total Visits: 1

Genetic Algorithm Optimization Toolbox [Update Link / Bad Link?](#)

GAOT implements simulated evolution in the MATLAB environment using both binary and real representations. Ordered base representation has also been added to the toolbox. This implementation is very flexible in the genetic operators, selection functions, termination functions as well as the evaluation functions that can be used. The implementation is described in a technical paper.

Submitted Jul 10, 1999
Updated Jan 23, 2004

Rating: N/A [Rate this link](#) Total Visits: 1

Some Links where have GAs discussion :

(2) <http://geneticalgorithms.ai-depot.com/>



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[AI Game Programming Wisdom](#)
Approximately 70 articles by Artificial Intelligence programmers discuss the skills and concepts needed to apply AI to game development.
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Economics and Artificial Intelligence James Kingsbery - Tuesday 01, April
Theories in economy are fundamentally based on assumptions. This article proposes using simulation to verify them, using computer controlled agents that model human behaviours according economists - experimentally validating or contradicting the theories.
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Paradigmatic Considerations for an AI Interface for a Wearable Computer Carol Stein - Saturday 15, March
AI interfaces can be annoying. How should the information be presented to humans? Two concerns which must be adequately addressed by any designer of a wearable computer's PC-based AI interface are distraction, and effect on human memory.
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essay



Getting Darwinian Evolution to Work Paul Almond - Wednesday 12, February
Suggests that Darwinian evolution has higher levels of abstraction and that explicitly exploiting these in attempts to evolve software on computers could allow programs of greater sophistication. A multi-layered system is proposed which uses the concept of 'evolving evolvability'.
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Collective Intelligence in Social Insects David Gordon - Tuesday 21, January
Biological systems have long been a source of inspiration for AI. The latest paradigm to make the virtual leap comes from social insects - ant colonies and swarms. This article gives an overview of promising technologies to emerge from this field, looking at reasons 'swarm thinking' might come to characterise the new millenium.
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AI: The Technological Trickster Jim Erkiletian - Saturday 26, October
Moving elegantly among social sciences, computational linguistics and classical AI disciplines, this article analyses the essence of computer thought compared to human reasoning, and emphasises its effects on society and future developments.
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Moderating Online Chat Rooms Michael T. Pagano - Thursday 24, October
This paper shows how self-organised systems appear among chaos, given a nurturing environment. Such a technique is employed for self-moderation of message boards. Three types of moderation are discussed: automatic filtering, hierarchical censoring, and self-moderation through bottom-up feedback.
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Machine Learning in Games Development Nick Palmer - Thursday 26, September

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(3) <http://www.aic.nrl.navy.mil/galist/>

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GA Archives

The Genetic Algorithms Archive

Please note: Many of the links in the GA Archive connect to sites outside of this NCARAI web server. We are not responsible for the content of these other sites.

The Genetic Algorithms Archive is a repository for information related to research in genetic algorithms and other forms of evolutionary computation. Available from this site are past issues of the GA-List (now called [EC Digest](#)), source code for many EA implementations, and announcements about EC-related conferences. Also, links are given to many interesting sites around the World with material related to evolutionary computation. This archive is maintained at [The Navy Center for Applied Research in Artificial Intelligence](#).

Contents:

- [Calendar of EC-related events](#)
- [GA-List Archive](#): back issues, source code, information
- [Links to other GA-related information](#)
- [Links to EC research groups' home pages](#)
- [NEW Dissertations on Evolutionary Computation](#)
- [Other related societies and sites](#)

Some Links where have GAs discussion :

(4) <http://www-illigal.ge.uiuc.edu/index.php3>



Illinois Genetic Algorithms Laboratory

<http://www-illigal.ge.uiuc.edu/index.php3>

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About IlliGAL And Genetic Algorithms

At the Illinois Genetic Algorithms Laboratory (IlliGAL), we study nature's search algorithm of choice, genetics and evolution, as a practical approach to solving difficult problems on a computer. Genetic algorithms (GAs) and evolutionary computation have...

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The Champaign Armory

The Armory was designed as a military drill hall, an athletic facility, and an assembly hall. The building is something of an engineering feat: the long span structure was unique in its day.

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A new book on [Hierarchical Bayesian Optimization Algorithm](#) by Martin Pelikan is now available.

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For information on the most recent technical reports and publications please see [What's New](#) or go directly to the [Technical Reports](#) page...

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Options	Source Code
FTP	XCS (+ tournament selection) classifier system implementation in C, version 1.2 (for IlliGAL Report 2003023)
FTP	Implementation of anticipatory classifier system (ACS2) in C++ (for IlliGAL Report 2001026)
FTP	Ordering Messy GA (OmeGA) in C++ (for IlliGAL Report 2000034)
FTP	Bayesian Optimization Algorithm with Decision Graphs in C++, version 1.1 (for IlliGAL Report 2000025)
FTP	XCS classifier system implementation in C, version 1.1
FTP	XCS implementation in Java, version 1.0 (for IlliGAL Report 2000017)
FTP	Bayesian Optimization Algorithm with Decision Graphs in C++, version 1.0 (for IlliGAL Report 2000025)
FTP	XCS classifier system implementation in C, version 1.0 (for IlliGAL Report 99021 .)
FTP	Extended Compact Genetic Algorithm in C++ (for IlliGAL Report 99016 .)
FTP	Bayesian Optimization Algorithm (BOA) code in C++ (for IlliGAL Report 99011 .)
FTP	Linkage Learning GA code in C++ (for IlliGAL Report 98006 .)
FTP	Messy GA code in C (for IlliGAL Report 91008 .)
FTP	Messy GA code in Lisp (for TCGA Report 90004 .)
FTP	Simple GA code (Pascal code from Goldberg, D. E. (1989), <i>Genetic Algorithms in Search, Optimization, and Machine Learning</i> .)
FTP	Simple GA code (C translation of the code from Goldberg, D. E. (1989), <i>Genetic Algorithms in Search, Optimization, and Machine Learning</i> .)
FTP	Simple Classifier System code (Pascal code from Goldberg, D. E. (1989), <i>Genetic Algorithms in Search, Optimization, and Machine Learning</i> .)

<http://www-illigal.ge.uiuc.edu/sourcecd.html>

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1	0	0	0	0	0	0	1	0	0	0	1	1	0	0	1
0	1	1	1	1	1	0	0	1	1	1	1	1	0	0	1
0	0	0	0	0	0	1	0	0	0	0	0	1	0	0	1
1	0	0	0	0	0	1	1	0	0	0	1	1	0	0	1
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SGALAB

Evolution with the world

SxLAB Family Member

What is Genetic Algorithms ?

Thanks.

End

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16-Jan-2009
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一张世界上智慧最集中的照片



1.彼得·德拜 美国物理化学家。1884年出生于荷兰。1901年进入德国亚琛工业大学学习电气工程，1905年获电子工程师学位，因他通过偶极矩研究及x射线衍射研究对分子结构学科所作贡献而于1936年获诺贝尔化学奖金。1966年逝世。

2.威廉·亨利·布喇格 (w.h.bragg, 1862 - 1942) 是现代固体物理学的奠基人之一，他早年在剑桥三一学院学习数学，曾任利兹大学、伦敦大学教授，1940年出任皇家学会会长。由于在使用x射线衍射研究晶体原子和分子结构方面所作出的开创性贡献，他与儿子w.l.布喇格分享了1915年诺贝尔物理学奖。父子两代同获一个诺贝尔奖，这在历史上恐怕是绝无仅有的。同时，他还作为一名杰出的社会活动家，在二三十年代是英国公共事务中的风云人物。

3. 爱因斯坦是20世纪最伟大的科学家，被公认为人类历史上最具有创造性才智的人物之一。他的名字与相对论密不可分，其实，相对论包括两种理论：其一是他1905年提出声狭义相对论；其二是他1915年提出的广义相对论。后者，我们最好称之为爱因斯坦引力论

4.埃伦费斯特 (p. ehrenfest, 1880 - 1933) ——荷兰物理学家

5.1930年，英国物理学家保罗·狄拉克 (paul adrien maurice dirac, 1902 ~ 1984) 用数学方法描述电子运动规律时，发现电子的电荷可以是负电荷、也可以是正电荷的。狄拉克猜想，在自然界中可能存在一种“反常的”带正电荷的电子

6.薛定谔 (erwin schrodinger, 1887-1961) 奥地利理论物理学家，与爱因斯坦、玻尔、玻恩、海森伯等一起于20世纪20年代后期，发展了量子力学。因建立描述电子和其他亚原子粒子的运动的波动方程，获得1933年诺贝尔物理奖。

7.1922—1923年间，康普敦 (a.h.compton 1892—1962) 研究了x射线经金属或石墨等物质散射后的光谱。

8.美籍奥地利科学家沃尔夫冈·泡利 (wolfgang e.pauli, 1900 ~ 1958)，是迎着20世纪一同来到世界的，父亲是维也纳大学的物理化学教授，教父是奥地利的物理学家兼哲学家。



9.海森伯, w.k. (werner karl heisenberg 1907~1976) 德国理论物理学家, 量子力学第一种有效形式(矩阵力学)的创建者。

10.玻恩, m. (max born 1882~1970) 德国理论物理学家, 量子力学的奠基人之一。

11. 尼尔斯·玻尔 (bohr, niels) 1885年10月7日生于丹麦首都哥本哈根, 父亲是哥本哈根大学的生理学教授。从小受到良好的家庭教育。1903年进入哥本哈根大学学习物理, 1909年获科学硕士学位, 1911年获博士学位。大学二年级时研究水的表面张力问题, 自制实验器材, 通过实验取得了精确的数据, 并在理论方面改进了物理学家瑞利的理论, 研究论文获得丹麦科学院的金奖章。

12. 普朗克, m. (max planck 1858~1947) 近代伟大的德国物理学家, 量子论的奠基人

13. 居里夫人 (1867-1934) 是最著名的女物理学家。她曾两次获诺贝尔奖, 1903年的物理奖, 1911年的化学奖。她受教育较晚, 于1893年获物理学位, 1894年获数学学位, 1903年获博士学位。居里夫人以放射性作为论文题目, 她研究了很多物质, 发现钍及其化合物的特性与铀相同。研究沥青铀矿时, 她发现了镭和钋。1910年她成功的分离了纯镭。居里夫人对巴黎的居里实验室的建立作出很大贡献。

14. 洛仑兹 (hendrik antoon lorentz 1853~1928) 与塞曼 (pietr zeeman 1865~1943) 因研究磁场对辐射现象的影响、发现塞曼效应, 分享了1902年度诺贝尔物理学奖。

15. 朗之万: 1872年1月23日生于巴黎, 法国著名的物理学家。



[念奴娇]

碧野黄花，蝶舞婀娜，桥头嬉逐绿鸭。燕雀未歇，催勤日，春曦始辞夜短。蹊径闲来，野旷天低，百草俯仰沁人。涂鸦纸上，素绘一片韵霞。

天墨蚀食初下，清风摆杨柳，星尘漫道。北斗转向，夜孤崖，古今云卷云舒。遥寄静月，江风曳渔火，山岚更暖。青竹更雅，水清浅伊入画。

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16-Jan-2009

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